

**BUSINESS REQUIREMENTS DOCUMENT
PWRAP - WRAP MATERIALS AND DIGITAL
PATTERNS MANAGEMENT SYSTEM**

GROUP 2

TUTORIAL CLASS: 04

VERSION: 3

DATE: 2026-04-16

VERSION AND APPROVALS

VERSION HISTORY			
Version #	Date	Revised By	Reason for change
0	9/3/2026	Vu Quynh Anh	Initial Definition of Project
1	12/3/2026	Vu Quynh Anh	UC Specification & Process Modeling
2	2/4/2026	Vu Quynh Anh	UC Specification & Process Modeling update and inclusion business case
3	16/4/2026	Vu Quynh Anh	Final version

Content changes between the current version and the previous version are identified using the Blackline convention (i.e., additions and ~~deletions~~).

This document has been approved as the official Business Requirements Document for **Pwrap - Wrap Materials and Digital Patterns Management System**, and accurately reflects the current understanding of business requirements. Following approval of this document, requirement changes will be governed by the project's change management process, including impact analysis, appropriate reviews and approvals.

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PROJECT DETAILS

Project Name	Pwrap - Wrap Materials and Digital Patterns Management System
Project Type	New Initiative
Project Start Date	09/03/2026
Project End Date	11/03/2026
Project Sponsor	Mr. Dao Duy Phong - CEO of Pwrap
Primary Driver	Efficiency
Secondary Driver	Asset Protection
Division	Tutorial Class 04 - Group 2
Project Manager	Dao Thanh Hien

OVERVIEW

This document defines the high level requirements **Pwrap - Wrap Materials and Digital Patterns Management System**. It will be used as the basis for the following activities:

- Creating solution designs
- Developing test plans, test scripts, and test cases
- Determining project completion
- Assessing project success

DOCUMENT RESOURCES

Name	Business Unit (Individual, Company or Department....)	Role	Proposed Elicitation Techniques
Mr. Dao Duy Phong	CEO of Pwrap	Sponsor	Interview

GLOSSARY OF TERMS

Term/Acronym	Definition
Wrap Material (Blank)	Uncut rolls of raw decal, tint film, or paint protection film (PPF) used for vehicles.
Digital Pattern (Cut File)	A vector-based design file (e.g., .EPS, .CDR) containing precise vehicle window or body panel dimensions, readable by cutting machines.
Pre-cut Wrap	The finished, ready-to-apply product after Wrap Materials are processed through a cutting machine using a Digital Pattern.
Vinyl Cutter	A computer-controlled cutting machine used to precisely cut Wrap Materials based on uploaded Digital Patterns.
Order Fulfillment	The complete process from the point of sales inquiry to delivery of a product (processing physical shipping or granting digital download access).
Inventory Stock	The real-time physical quantity of Wrap Materials available in the Pwrap warehouse.
Payment Gateway	A third-party service (e.g., VNPay, MoMo, Stripe) integrated into the system to authorize and process online payments for orders.
Wholesale Customer (Dealer)	Auto shops or detailing centers that purchase bulk Wrap Materials and access Digital Patterns for their own business operations.

PROJECT OVERVIEW

4.1 Project Overview and Background

This project focuses on the development and implementation of "Pcut" – a centralized internal management system specifically designed for Pwrap.vn, a reputable retailer in the automotive wrap and skin industry. The system is engineered to synchronize the management of physical wrap materials with the company's high-precision digital cut files. Currently, the business operates with fragmented processes, where physical stock and digital assets are managed separately. This disconnection often leads to inventory inaccuracies, operational delays, and a significant risk to the company's intellectual property.

The main problem addressed by this project is the lack of a secure, integrated platform to track physical inventory and safeguard proprietary digital patterns. Manual or uncoordinated methods of monitoring wrap material stocks are not only time-consuming but also prone to errors. Furthermore, without a centralized security layer, the company's unique digital cut files, its core competitive advantage, are highly vulnerable to unauthorized access and external leaks, which could severely impact Pwrap.vn's market position.

The primary objectives of the Pcut project are:

1. **Centralized Inventory Tracking:** To provide a unified platform for real-time monitoring of physical wrap material stocks, ensuring accurate data on available blank materials for production.
2. **Digital Asset Protection and Security:** To implement a secure environment with strict access controls to protect proprietary digital cut files from unauthorized copying, theft, or external leaks.

- 3. Automated Workflow Integration: To streamline the process of matching digital patterns with physical materials, reducing manual handling, and optimizing overall business operations.
- 4. Resource Optimization: To minimize material waste and administrative burden by providing precise stock levels and efficient digital file retrieval.
- 5. Role-Based Access Management: To establish clear permissions for internal staff, ensuring that employees can access necessary files for daily tasks without compromising sensitive company data.
- 6. Operational Reliability: To enhance the speed and accuracy of the customer fulfillment process, thereby improving business scaling capabilities and customer satisfaction.

4.2 Project Dependencies

N/A

4.3 Stakeholders

The following comprises the internal and external stakeholders whose requirements are represented by this document:

	Stakeholders
1.	Admin (Internal)
2.	Staff (Internal)
3.	Dealer (External)

(OPTIONAL) KEY ASSUMPTIONS AND CONSTRAINTS

5.1 Key Assumptions and Constraints

#	Assumptions
#	Constraints

BUSINESS REQUIREMENTS

The following sections document the various business requirements of this project.

The requirements in this document are prioritized as follows:

Value	Rating	Description
1	Critical	This requirement is critical to the success of the project. The project will not be possible without this requirement.
2	High	This requirement is high priority, but the project can be implemented at a bare minimum without this requirement.
3	Medium	This requirement is somewhat important, as it provides some value but the project can proceed without it.
4	Low	This is a low priority requirement, or a “nice to have” feature, if time and cost allow it.
5	Future	This requirement is out of scope for this project, and has been included here for a possible future release.

Functional Requirements

REQ#	PRIORITY	DESCRIPTION	RATIONALE	USE CASE
F01	1	Users should be able to log in to the system using their Email/Username and Password.	Protects the internal system from unauthorized access.	1. Log in to the system
F02	1	Admins should be able to grant granular module access (Files, Inventory, Orders) to Staff accounts based on specific permissions rather than fixed roles.	Maximizes resource optimization by allowing a staff member to both create orders and manage inventory, provided the Admin grants both permissions to the same account.	2. Manage Staff permissions
F03	1	Users should be able to maintain only one active session at a time; logging into a second device or browser will automatically terminate the first session (Concurrent Login Prevention).	Supports incident tracing, security audits, and identifying the source of data leaks or inventory discrepancies.	3. View system logs
F04	1	Admins should be able to upload vector files (.cdr, .eps, .dxf) along with required metadata (Car Make, Model, Year, Cut Details).	Builds the core digital library	4.Upload cut file
F05	1	Users should be able to search files by keyword and filter results by Car Make, Model, Year, and Material Type (PPF, Color Wrap).	Technicians need to find files quickly to start cutting and optimizing customer wait times.	5.Search cut file
F06	1	Staff should be able to restock new rolls of wrap/PPF with details: Material Code, Color, Size (linear meters), Supplier, and Batch Number.	Keeps inbound stock levels accurate.	6.Restock materials

F07	1	Admin / Staff should be able to view the current stock levels of each material code (in rolls or linear meters) on the dashboard.	Provides Sales with accurate data to advise customers on material availability.	7.View inventory
F08	2	Admin / Staff should be able to receive alerts and see highlights when a material code drops below a set threshold (e.g., < 15 meters).	Helps procurement order materials in time to prevent supply chain disruptions.	8.View inventory alerts
F09	1	Staff should be able to create orders to sell blank materials or cut files to authorized dealers, which includes: Select Customer, Select Product, Enter Quantity, Apply Discount.	Professionally scales Pwrap's wholesale operations.	9.Create order
F10	2	Admin should be able to upload a new version of a cut file while retaining the previous versions in the database history.	Allows the business to continuously optimize cut patterns without losing historical data or accidentally overriding functional files with flawed updates.	10.Update cut file version
F11	2	Staff should be able to view an automatically generated low-resolution, watermarked image preview of the digital cut file before deciding to download it.	Technicians can visually verify if it's the correct pattern without downloading the actual vector file, reducing unnecessary downloads and minimizing exposure.	11.Preview cut file
F12	2	Admin/Staff should be able to view and track the real-time status of any B2B Dealer's order (e.g., from Awaiting Payment to Shipping) directly through the system's management dashboard.	Improves order management efficiency, enabling internal teams to proactively oversee the delivery progress of wrap materials, anticipate potential delays, and provide prompt support to B2B Dealers..	12.Track order status
F13	2	Admin / Staff should be able to process returned B2B orders (e.g., defective materials), which allows the system to automatically restore the physical inventory count and log the return reason.	Physical warehouses always face the risk of returned goods. The system must record this reverse inventory flow to prevent data discrepancies.	13.Process material return
F14	2	Admin / Staff should be able to export system data such as inventory status, download logs, and B2B sales reports into Excel/CSV formats for specified date ranges.	Facilitates accounting, internal audits, and monthly business analysis operations for Pwrap.	14.Generate System Reports
F15	1	Admin should be able to add a new product to the catalog by inputting details such as Product Name, SKU/Product Code, Category (e.g., Material Roll, Pre-cut Kit, Tool), Base Price, and uploading product images.	Creates the product Master Data, a foundational requirement that enables Sales personnel to select and include products in newly created orders.	15.Add product
F16	2	Admin should be able to update existing product information, including modifying the Base Price, changing descriptions, or toggling the product visibility status	Allows the business to flexibly adjust selling prices based on market conditions or update product details	16.Update products information

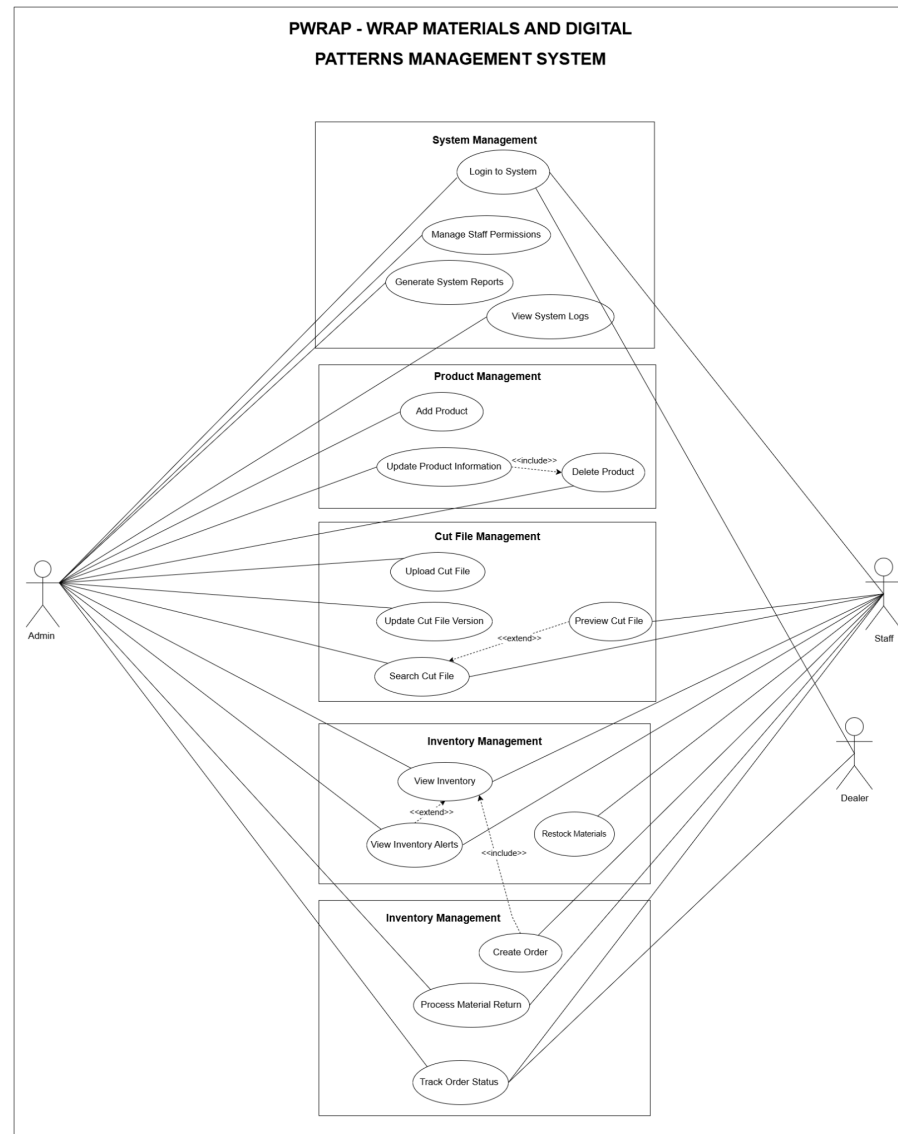
Non-functional Requirements

REQ#	PRIORITY	DESCRIPTION	RATIONALE	USE CASE
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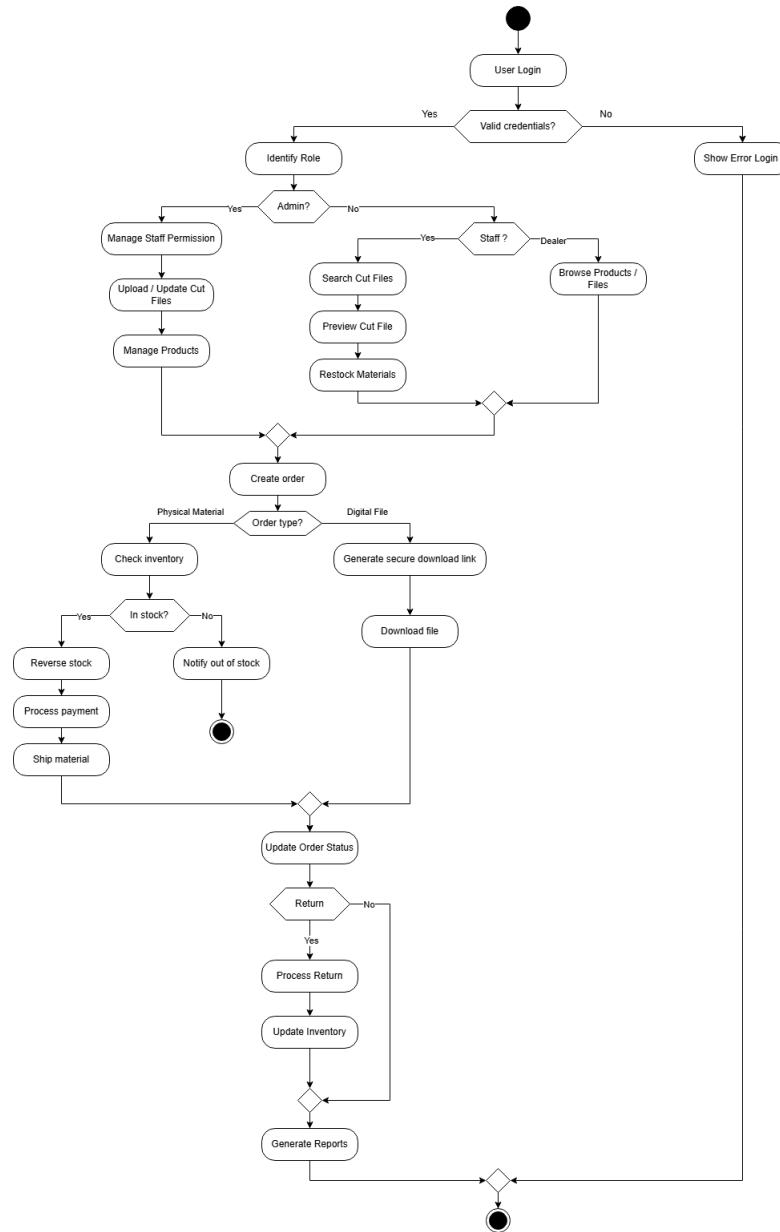
NF01	2	The system shall support at least 500 concurrent users without significant performance degradation.	Ensures system stability during peak usage periods.	1.Login to System
NF02	2	The system shall process the upload of new cut files (≤50MB) within 5 seconds.	Allows administrators to update the cut-file database efficiently	4.Upload Cut File
NF03	2	Search results for cut files based on car model or window type shall be displayed within 2 seconds.	Enables users to quickly locate the required cut file.	5.Search Cut File
NF04	1	The system shall provide temporary download tokens or expiring links to prevent unauthorized sharing of cut files.	Reduces the risk of illegal distribution of proprietary cut files.	17.Download Cut File
NF05	1	The system response time for common actions (loading pages, viewing products, searching files) shall not exceed 2 seconds.	Improves usability and efficiency for both customers and staff.	18.View Product

USE CASES

Use Case Diagram



Process Diagram



Use Case Specification

ID & Name:	UC-02: Management Staff Permissions
Primary Actor:	Admin
Description:	This use case allows the admin to assign, modify, r revoke system permissions for staff members to control their access system functions such as product management, inventory management, and cut file management.
Trigger:	Admin selects the Manage Staff Permissions function from the system management menu.
Pre-conditions:	Admin must be logged into the system. Staff accounts mst already exist in the system.
Post-conditions:	Staff permissions are updated successfully. The system records the permission changes in the system log.
Normal Flow:	1.1. Admin logs into the system. 1.2. Admin navigates to System Management. 1.3. Admin selects Manage Staff Permissions. 1.4. The system displays the list of staff account. 1.5. Admin select a specific staff member. 1.6. The system displays current permissions of the selected staff. 1.7. Admin assigns or modifies permission. 1.8. Admin confirms the changes. 1.9. The system saves the updated permissions. 1.10. The system displays a confirmation message.
Alternative Flows:	2.1. Admin searches for a staff member. 2.2. The system cannot find the account. 2.3. The system displays "Staff account not found". 2.4. Admin starts editing permissions. 2.5. Admin select Cancel. 2.6. The system discards all changes.
Exceptions:	Exception #1: System error while saving: 1.2 Admin confirms permission changes 1.3 System fails to save the data 1.4 The system displays an error message. 1.5 Admin is asked to retry.
Priority:	High

Use Case Graphic (if any)

ID & Name:	UC-03: View System Logs
Primary Actor:	Admin
Description:	An Admin views system logs to trace incidents, perform security audits, and identify sources of data leaks or inventory discrepancies.
Trigger:	Admin clicks on the 'System Logs' link in the management dashboard.
Pre-conditions:	The User is logged in. The User has the Admin role.
Post-conditions:	System logs are displayed on the page for the Admin to review.
Normal Flow:	1.1 Displays list of recent system events (e.g., login events, session terminations, file downloads). 1.2. Admin inputs criteria into search/filter fields (e.g., specific Staff ID, date range, or "concurrent login" events). 1.3. Clicks 'Filter'. 1.4. System updates the displayed log records to match the filter criteria.
Alternative Flows:	2.1. Admin clicks 'Export Logs'. 2.2. System prompts for a date range and file format (Excel/CSV). 2.3. Admin selects criteria and clicks 'Download'. 2.4. System generates and downloads the log report.
Exceptions:	Exception #1: No logs match search criteria 1.2. System displays "No records found matching your criteria."
Priority:	High

Use Case Graphic (if any)

ID & Name:	UC-04: Upload Cut File
Primary Actor:	Staff / Admin
Description:	This use case allows authorized staff or administrators to upload new digital cut file into the system for storage and future use in the car wrap cutting process.
Trigger:	User selects Upload Cut File from Cut File Management module.
Pre-conditions:	User must be logged into the system. User must have permissions to upload files. The file must be in a supported format.
Post-conditions:	The cut file is successfully uploaded and stored in the system database. The file becomes searchable and accessible for future use.
Normal Flow:	1.1. User logs into the system. 1.2. User navigates to Cut File Management. 1.3. User selects Upload Cut File. 1.4. The system displays the file upload interface. 1.5. User selects the cut file from the local device. 1.6. User enter file information (car model, window type, version). 1.7. User clicks Upload. 1.8. The system validates the file format and size. 1.9. The system uploads and stores the file. 1.10. The system confirms the succesful uploat.
Alternative Flows:	2.1. User select a file with an unsupported format. 2.2. The system displays "Invalid file format". 2.3. User selects another file. 2.4. User attempts to upload the file without entering required data. 2.5. The system displays a warning. 2.6. User completes the required fields
Exceptions:	Exception #1: Upload failure: 1.2 System fails during the upload process. 1.3 System displays Upload failed message. 1.4 User is asked to retry.
Priority:	High

Use Case Graphic (if any)

ID & Name:	UC-05: Search Cut File
Primary Actor:	Staff, Admin
Description:	Search for vector file in the technical library based on the customer's vehicle information.
Trigger:	User accesses the "Digital Library" module and interacts with the search bar.
Pre-conditions:	User is logged in to the system; User is authenticated and authorized to access technical assets.
Post-conditions:	A list of matching files is displayed for the user to select, search history is optionally logged.
Normal Flow:	1.1. User selects the vehicle's Make (e.g., Honda, Toyota) from the dropdown list. 1.2. User selects the specific Model and Year of Manufacture. 1.3. User enters additional keywords (eg., "Door", "Hood") into the search bar. 1.4. User clicks the "Search" button to execute the query. 1.5. System validates the input criteria and queries the technical database. 1.6. System displays a list of matching vector files with metadata
Alternative Flows:	2.1. View file sample: User click on a specific file to view a watermark Preview image before selecting. 2.2. Clear Filters: User clicks "Reset" to clear all current search parameters and start a new query.
Exceptions:	Exception #1: No result found 1.2. System queries the database. 1.3. System displays "No matches found for the criteria!" Exception #2: Database Timeout 2.2. System queries the database.

	2.3. System displays "Connection timeout. Please refresh and try again!"
Priority:	High
Use Case Graphic (if any)	

ID & Name:	UC-06: Restock materials
Primary Actor:	Staff
Description:	Staff updates the system inventory by adding newly received physical wrap materials, recording details such as Material Code, Color, Size, Supplier, and Batch Number.
Trigger:	Staff clicks the "Restock Material" button.
Pre-conditions:	The User is logged in. The User has the Staff role. The material profile exists in the system database.
Post-conditions:	The material's inventory quantity is updated and a restock transaction log is saved.
Normal Flow:	1.1. Staff searches and selects the Material Code. 1.2. Staff clicks "Restock". 1.3. Staff inputs restock details (Quantity, Color, Size, Supplier, Batch). 1.4. Staff clicks "Confirm". 1.5. System validates the input data. 1.6. System updates the total inventory quantity. 1.7. System saves the transaction log. 1.8. System displays "Restock Successful".
Alternative Flows:	None
Exceptions:	Exception #1: Invalid Quantity 1.2. Staff inputs an invalid Quantity (0 or negative). 1.3. Staff clicks "Confirm". 1.4. System displays "Error: Quantity must be greater than zero!" and prompts User to re-enter. Exception #2: Missing mandatory fields 2.2. Staff leaves mandatory fields empty (e.g., Supplier or Batch). 2.3. Staff clicks "Confirm". 2.4. System displays "Error: All fields (Color, Size, Supplier, Batch) are required!" and prompts User to re-enter.

Priority:	High
Use Case Graphic (if any)	

ID & Name:	UC-07: View Inventory
Primary Actor:	Staff, Admin
Description:	Allows Staff, Admin to view the list of materials in the inventory, including quantity and status.
Trigger:	When Staff, Admin click on the "View inventory" button on the Inventory Management page
Pre-conditions:	Staff/Admin is logged in into the system
Post-conditions:	The system displays the current inventory information
Normal Flow:	1.1. Staff/Admin scrolls in the Inventory list 1.2. Staff/Admin 1.3. Staff/Admin reviews the material information
Alternative Flows:	2.1. Staff/Admin enters material name or ID in the search box 2.2. Staff/Admin selects the item in the search results 2.3. Staff/Admin reviews the material information
Exceptions:	Exception #1: Inventory data cannot be retrieved 1.2. System attempts to load inventory data 1.3. System displays: "Error: Unable to retrieve inventory data."
Priority:	High
Use Case Graphic (if any)	

ID & Name:	UC-09: Create Order
Primary Actor:	Staff
Description:	Staff create an order for a dealer (existing or new). The system calculates the price and reserves stock.
Trigger:	Staff click the "Create Order" button on the order management interface.
Pre-conditions:	Staff is logged in; The system has access to inventory and customer data.
Post-conditions:	Order is saved with "Draft" status; Physical materials are reserved in stock.
Normal Flow:	1.1. Staff clicks "Create Order" 1.2. System opens the order creation form. 1.3. Staff searches for and selects a dealer from existing list. 1.4. Staff adds physical materials and enters the quantity (linear meters) 1.5. System checks inventory and automatically applies the discount. 1.6 Staff clicks "Submit" to complete the order.
Alternative Flows:	2.1. Dealer not in the list: Staff selects "Add New Dealer" and enter information of Dealer 2.2. Select Digital Cut File: Staff select "Cut File" instead of "Physical Materials", System automatically retrieves the storage path instead of deducting linear meters from stock. 2.3. Save as Draft
Exceptions:	Exception #1: Out of Stock 1.2. System check inventory stock. 1.3. System displays "Insufficient Stock for requested amount!" Exception #2: Overdue dept 2.2. Click submit 2.3. System displays "Dealer has overdue balance. Admin approval required"
Priority:	Critical
Use Case Graphic (if any)	

ID & Name:	UC-10: Update Cut File Version
Primary Actor:	Admin
Description:	An Admin uploads a new version of a cut file while retaining the previous versions in the database history to continuously optimize patterns.
Trigger:	Admin clicks the 'Update Version' button on the detail page of an existing cut file.
Pre-conditions:	The Admin is logged in. The cut file already exists in the system.
Post-conditions:	The new cut file version is saved and set as the active download, while the older version is securely archived in the database history.
Normal Flow:	1.1. Admin accesses the detail page of a specific cut file. 1.2. Admin clicks the 'Update Version' button. 1.3. System shows an 'Upload New Version' form. 1.4. Admin uploads the file that has valid types (.cdr, .eps, or .dxf) and inputs optional version notes. 1.5. Admin clicks 'Submit'. 1.6. System processes the upload within 5 seconds. 1.7. System saves the new file as the active version, moves the old file to version history, and displays "Update Successful!"
Alternative Flows:	2.1. Admin clicks 'Update Version'. 2.2. Admin views the 'Upload New Version' form. 2.3. Admin clicks 'Cancel'. 2.4. System aborts the update and returns to the file detail page.
Exceptions:	Exception #1: File size limit exceeded or file format wrong 1.2. System displays "Error: File size exceeds the 50MB limit or file limit wrong."
Priority	High
Use Case Graphic (if any)	

ID & Name:	UC-12: Track order status
Primary Actor:	Staff, Admin
Description:	Admin or Staff views and monitors the real-time status of B2B Dealer orders through the system dashboard to track the order progress (Awaiting Payment, Processing, Shipping, Completed)
Trigger:	Admin/Staff clicks the "orders" or "Track order status" menu on the dashboard.
Pre-conditions:	The user is logged into the system. The user has Admin or Staff permissions. At least one order exists in the system.
Post-conditions:	The selected order status is displayed. Admin/Staff can monitor the progress of the order.
Normal Flow:	1.1. Admin/Staff opens the Order Management Dashboard. 1.2. Admin/Staff selects an order from the order list. 1.3. System displays the detailed order information. 1.4. System shows the current order status (Awaiting Payment, Processing, Shipping, Completed). 1.5. Admin/Staff reviews the order progress.
Alternative Flows:	2.1. Admin/Staff searches for an order using order ID or Dealer name. 2.2. System filters and displays the matching orders. 2.3. Admin/Staff selects a specific order. 2.4. System displays the order status.
Exceptions:	Exception #1: Order not found 1.2. Admin/Staff searches for an order. 1.3. System displays "Error: Order not found!" Exception #2: System cannot retrieve order status 2.2 System fails to load order information. 2.3. System displays "Error: Unable to retrieve order status. Please try again later."
Priority:	High
Use Case Graphic (if any)	

ID & Name:	UC-13: Process Material Return
Primary Actor:	Staff, Admin
Description:	Allow Staff/Admin to process the returned of materials from an order. The system updates the inventory and record the return transaction
Trigger:	Staff/Admin selects the "Process Material Return" function from the Order Management page
Pre-conditions:	Staff/Admin is logged in into the system The order already exists in the system The order contains materials that can be returned
Post-conditions:	The returned materials are recorded in the system Inventory quantity is updated The return transaction is saved for tracking
Normal Flow:	1.1. Staff/Admin searches or selects the order that contains the materials to return 1.2. Staff/Admin selects the material to return 1.3. Staff/Admin enters the return quantity 1.4. Staff/Admin enters 1.5. Staff/Admin confirms the return request
Alternative Flows:	2.1. Staff/Admin selects the materials to return 2.2. Staff/Admin enters the quantity 2.3. Staff/Admin enters the reason for return 2.4. Staff/Admin reviews the return information 2.5. Staff/Admin modifies the return quantity or reason 2.6. Staff/Admin confirms the return request
Exceptions:	Exception #1: Invalid Return Quantity 1.2. Staff/Admin enters a return quantity greater than the original order quantity 1.3. System displays an error message: "The quantity is invalid."
Priority:	High
Use Case Graphic (if any)	

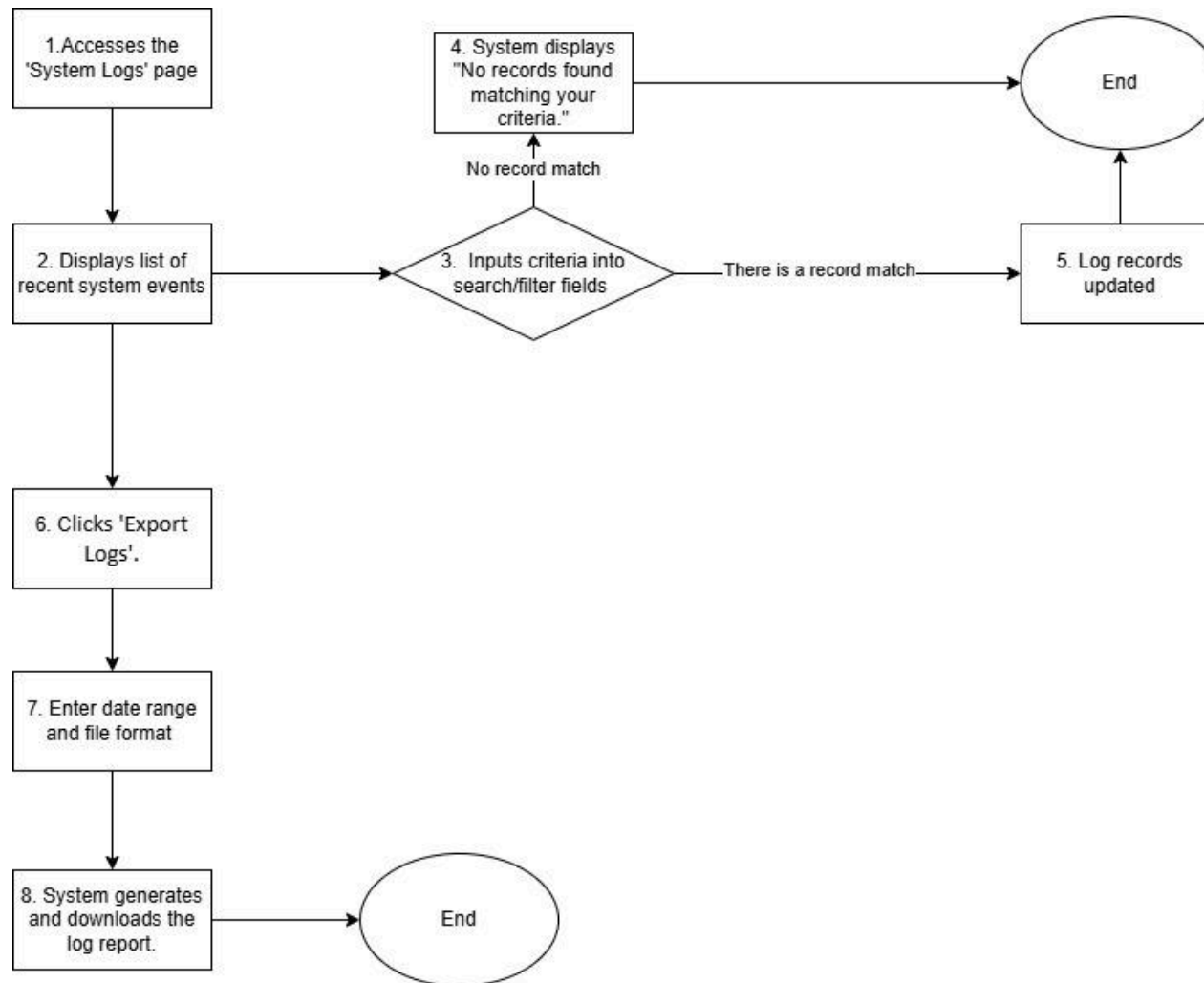
ID & Name:	UC-14: Generate System Reports
Primary Actor:	Admin / Staff
Description:	Admin or Staff exports system data (inventory status, download logs, and B2B sales reports) into Excel or CSV formats for specified date ranges.
Trigger:	Admin / Staff clicks on the "Reports" menu item.
Pre-conditions:	The User is logged in. The User has the Admin or Staff role. The system database is accessible.
Post-conditions:	A report file is generated and successfully downloaded to the User's device.
Normal Flow:	1.1. Admin / Staff selects the report type (inventory status, download logs, or B2B sales). 1.2. Admin / Staff inputs the Date Range (Start Date, End Date). 1.3. Admin / Staff clicks "Export to Excel". 1.4. System validates the date range. 1.5. System retrieves matching data from the database. 1.6. System generates and downloads the Excel file. 1.7. System displays "Export Successful".
Alternative Flows:	2.1. System validates the date range. 2.2. System retrieves data, generates and downloads the CSV file. 2.3. System displays "Export Successful".
Exceptions:	Exception #1: Invalid Date Range 1.2. Admin / Staff inputs an invalid Date Range (End Date is before Start Date). 1.3. Admin / Staff clicks "Export". 1.4. System displays "Error: End Date must be after Start Date!" and prompts User to re-enter. Exception #2: No data found 2.2. System retrieves data but finds 0 records. 2.3. System displays "Notification: No data available for the selected period!"
Priority:	High
Use Case Graphic (if any)	

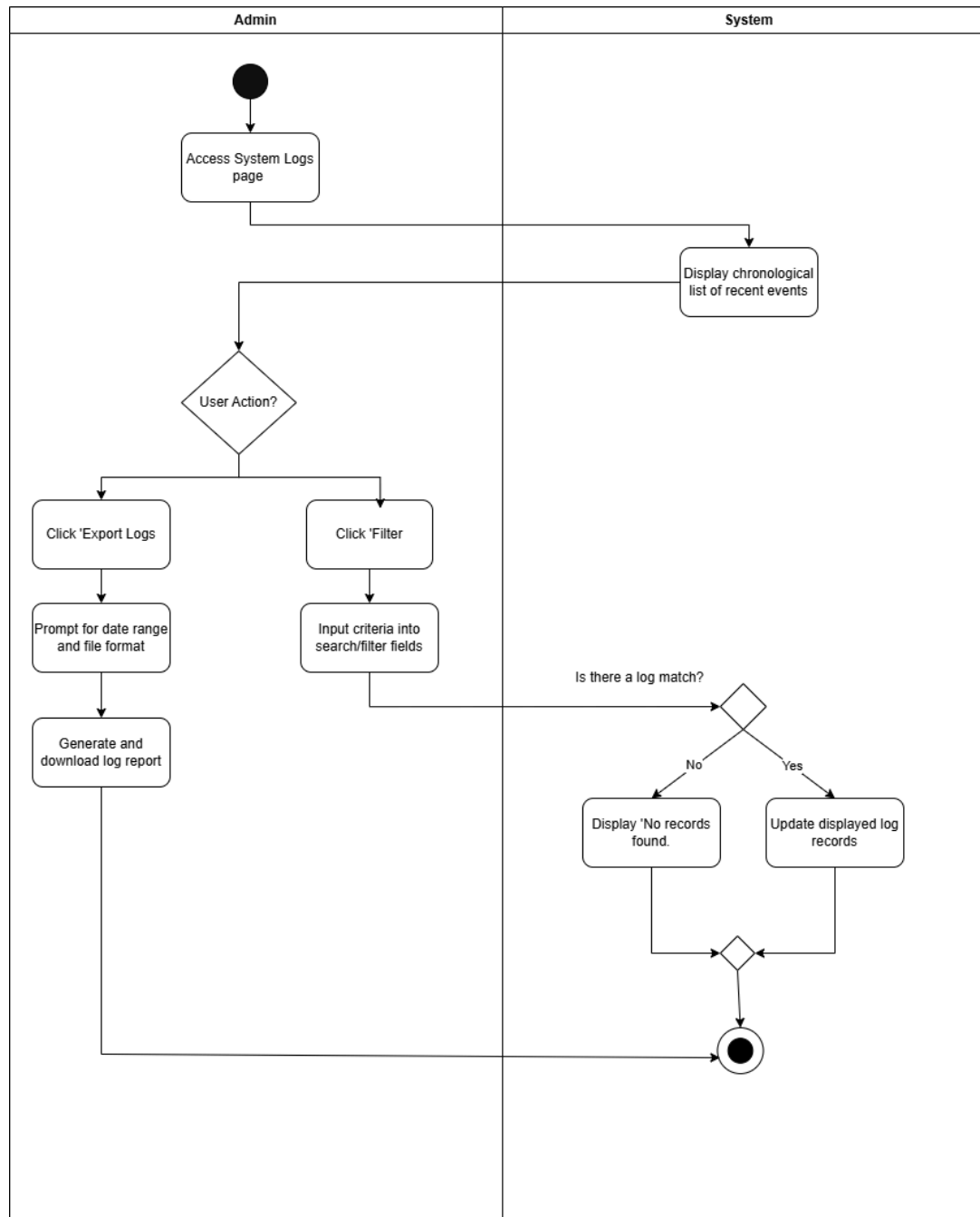
ID & Name:	UC-15: Add product
Primary Actor:	Admin
Description:	Admin adds a new product to the catalog by inputting details such as Product Name, Product Code, Category (Material Roll, Pre-cut Kit, Tool), Base Price, and uploading product images
Trigger:	Admin clicks the "Add Product" button.
Pre-conditions:	The user is logged into the system The user has Admin permissions
Post-conditions:	The product Master Data is created, enabling Staff personnel to select and include products in newly created orders
Normal Flow:	1.1 Admin clicks "Add Product". 1.2 System displays the Add Product form. 1.3 Admin inputs product details such as Product Name, SKU/Product Code, Category, and Base Price. 1.4 Admin uploads product images. 1.5 Admin clicks submit. 1.6 System validates the information. 1.7 System displays "Successful".
Alternative Flows:	2.1. Admin inputs product details 2.2. Admin clicks cancel. 2.3. System clears the form and returns to the previous page.
Exceptions:	Exception #1: Product Code is duplicated. 1.2. Admin clicks submit. 1.3. System displays "Error: Duplicated Product Code!" Exception #2: Missing mandatory information. 2.2. Admin clicks submit. 2.3. System displays "Error: Please fill in all required fields!".
Priority:	Critical
Use Case Graphic (if any)	

APPENDIXES

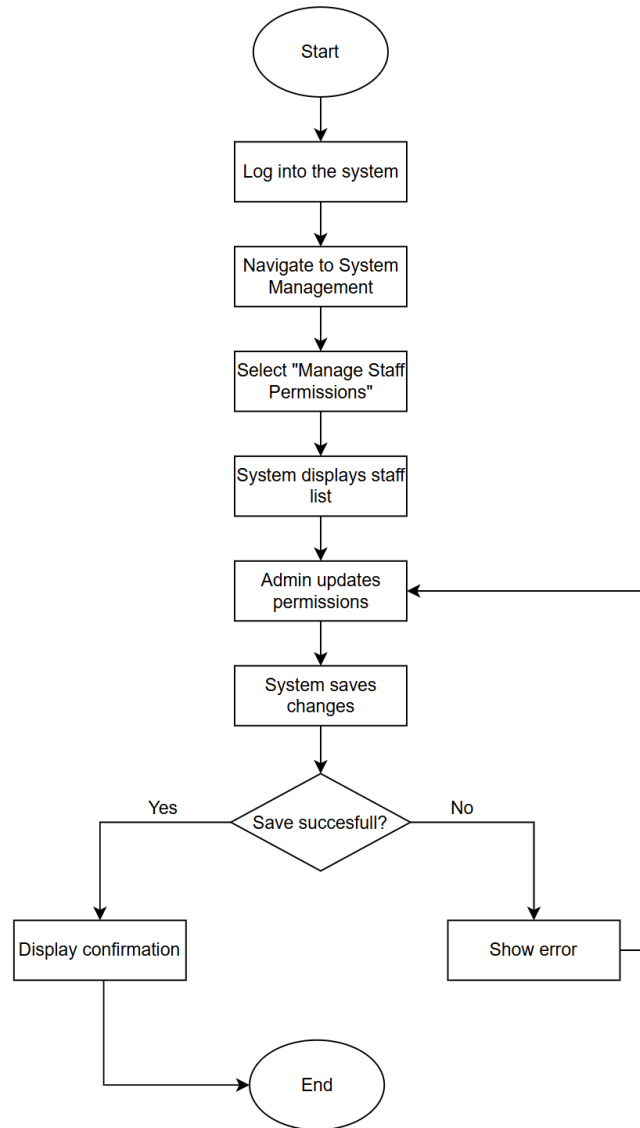
Appendix A – Business Process Flows

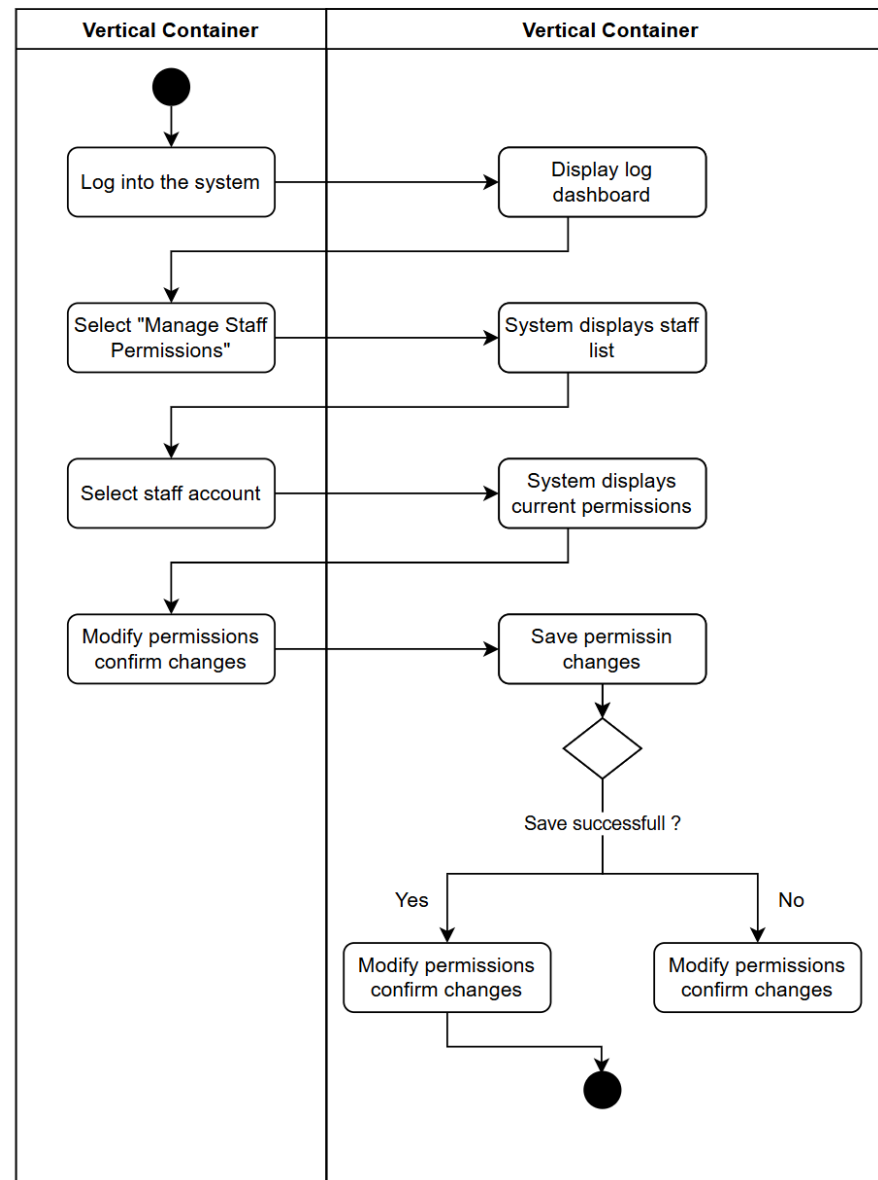
Use Case 01:



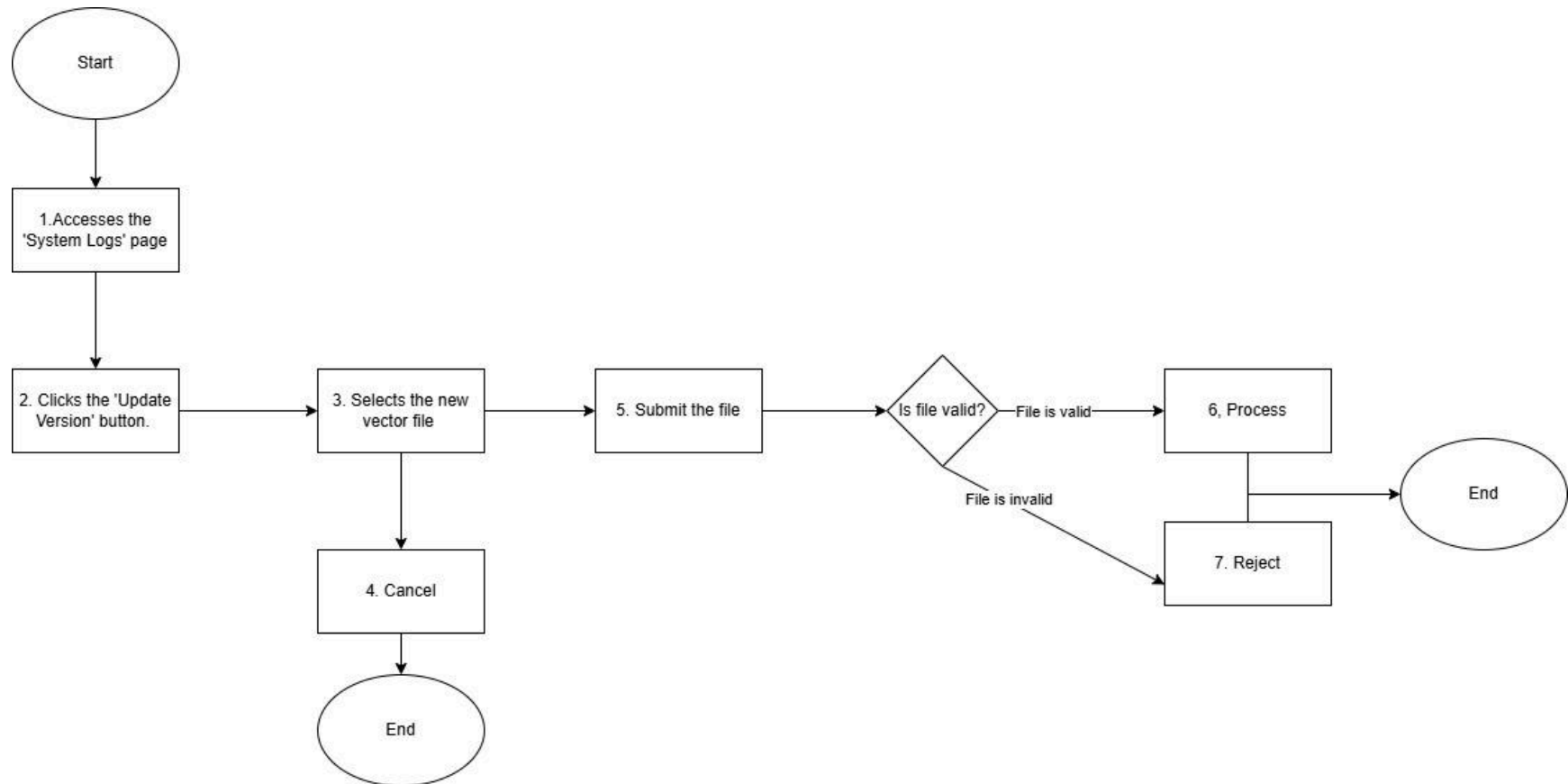


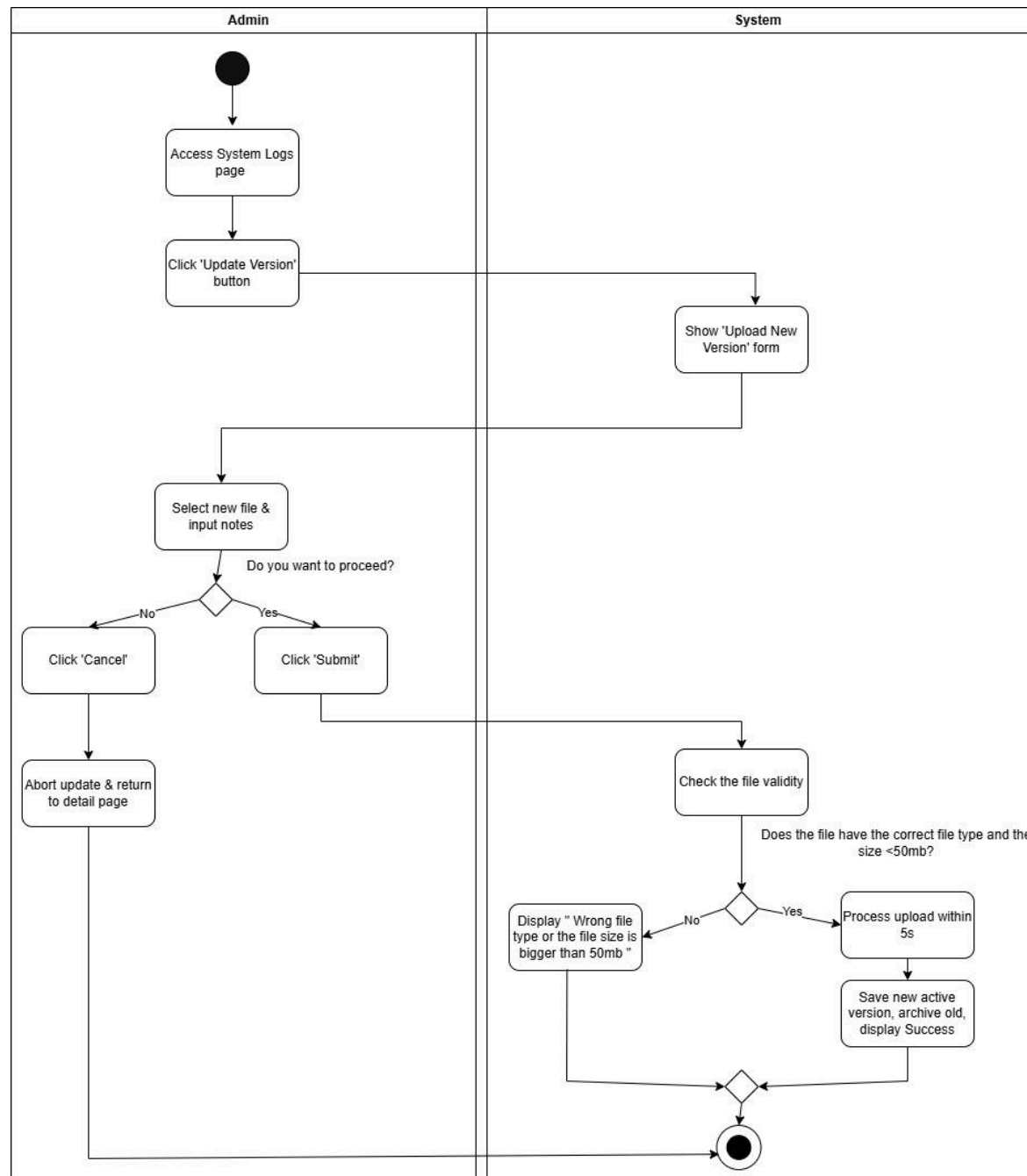
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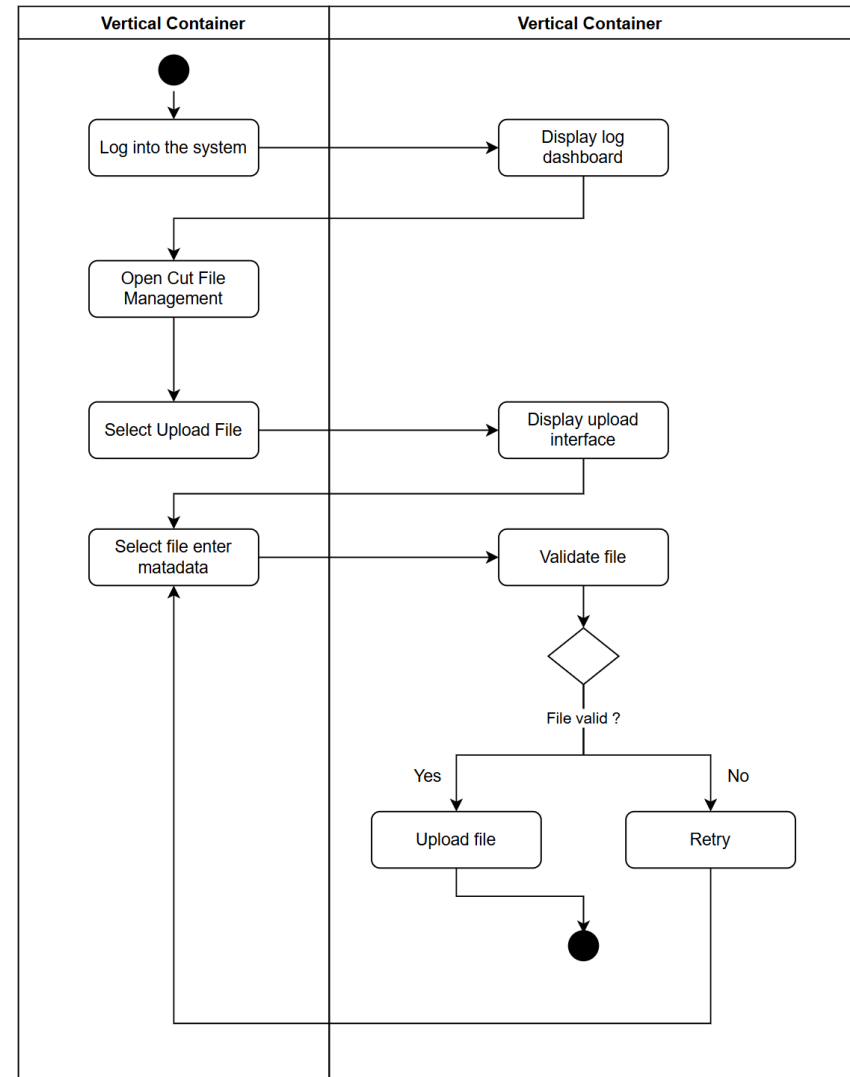
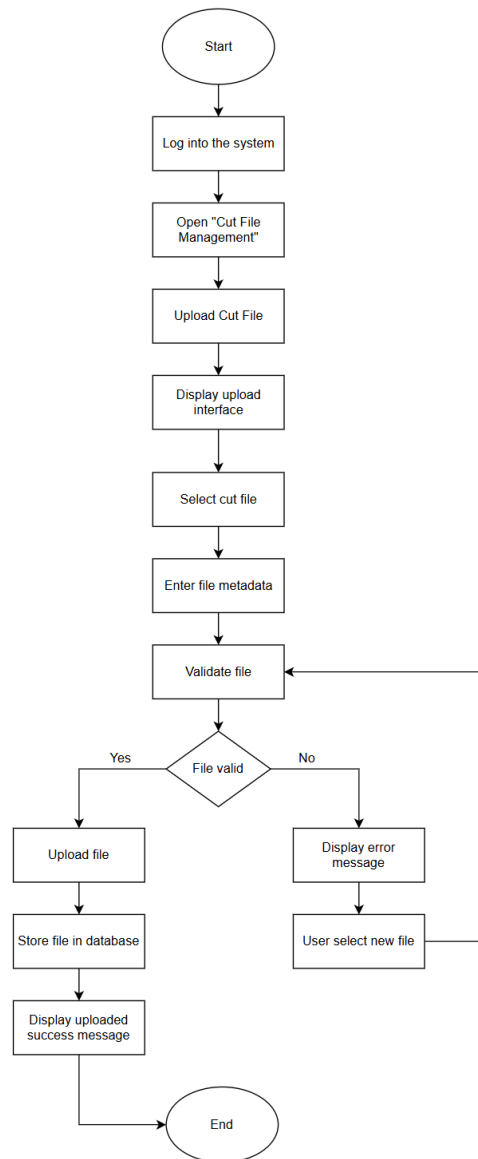


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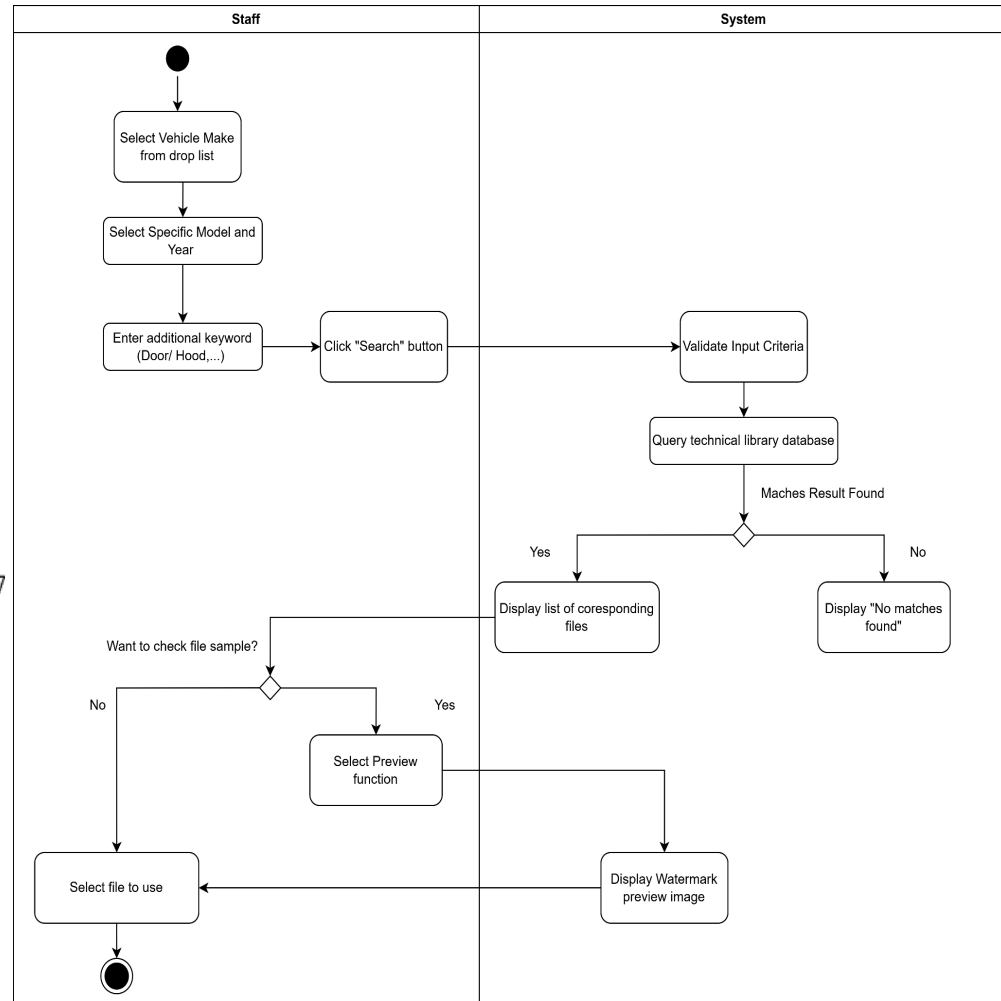
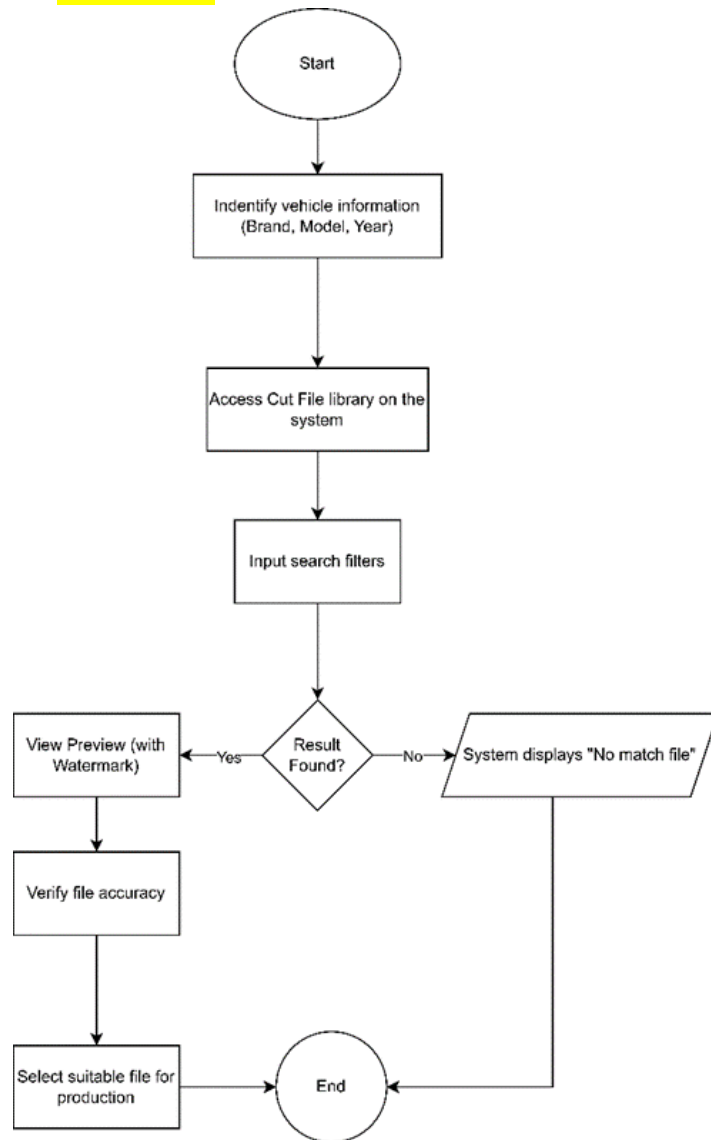




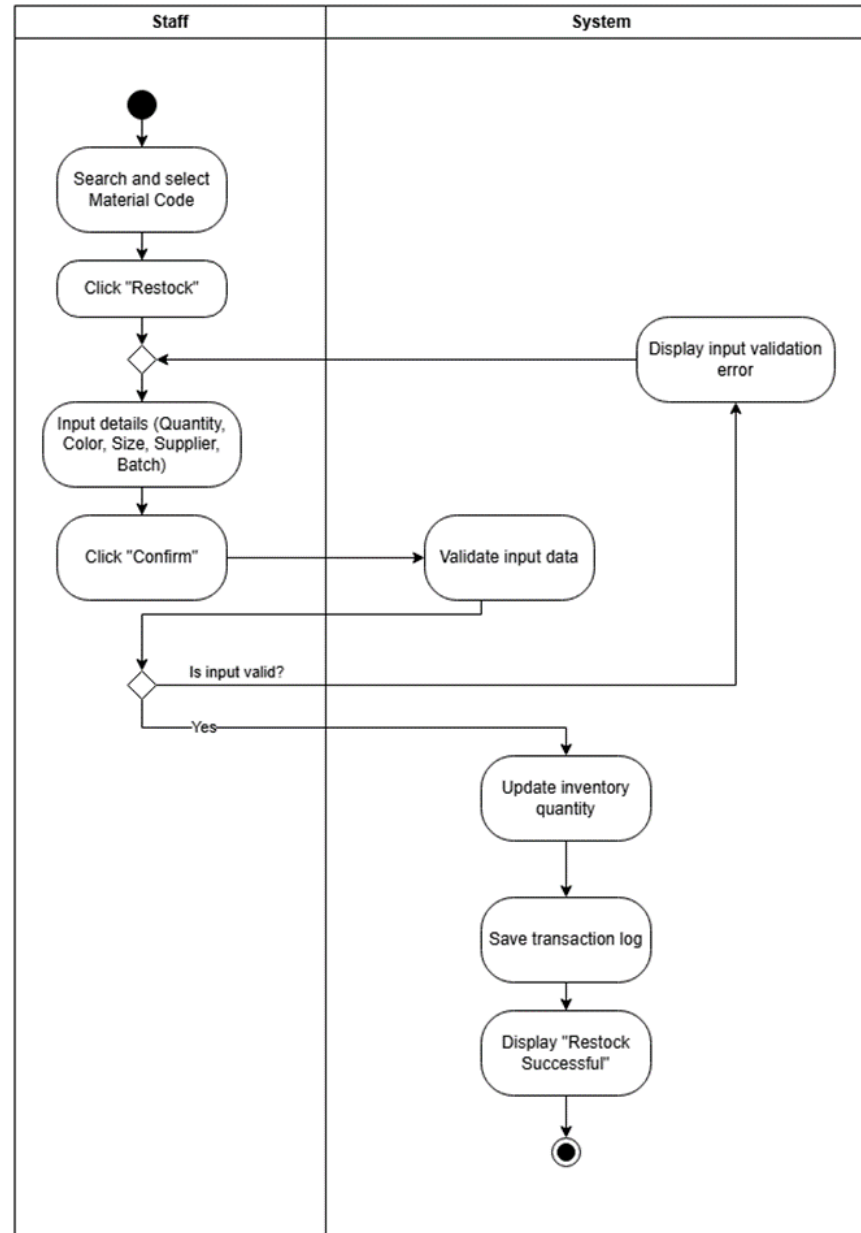
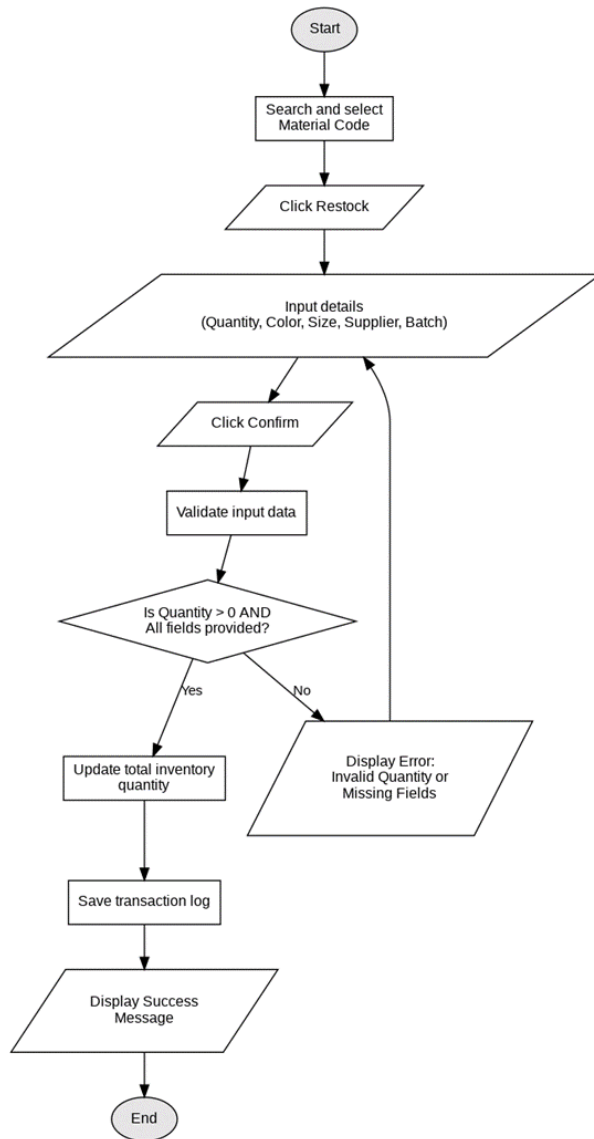
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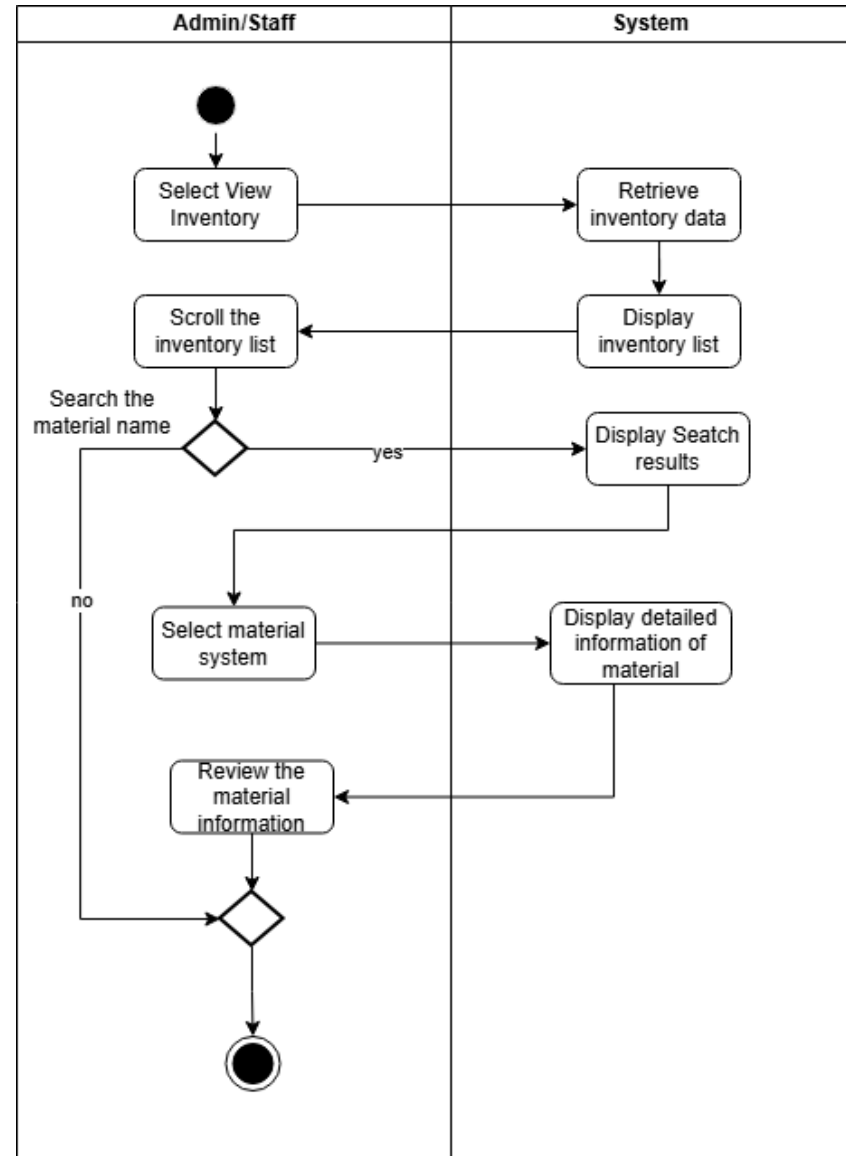
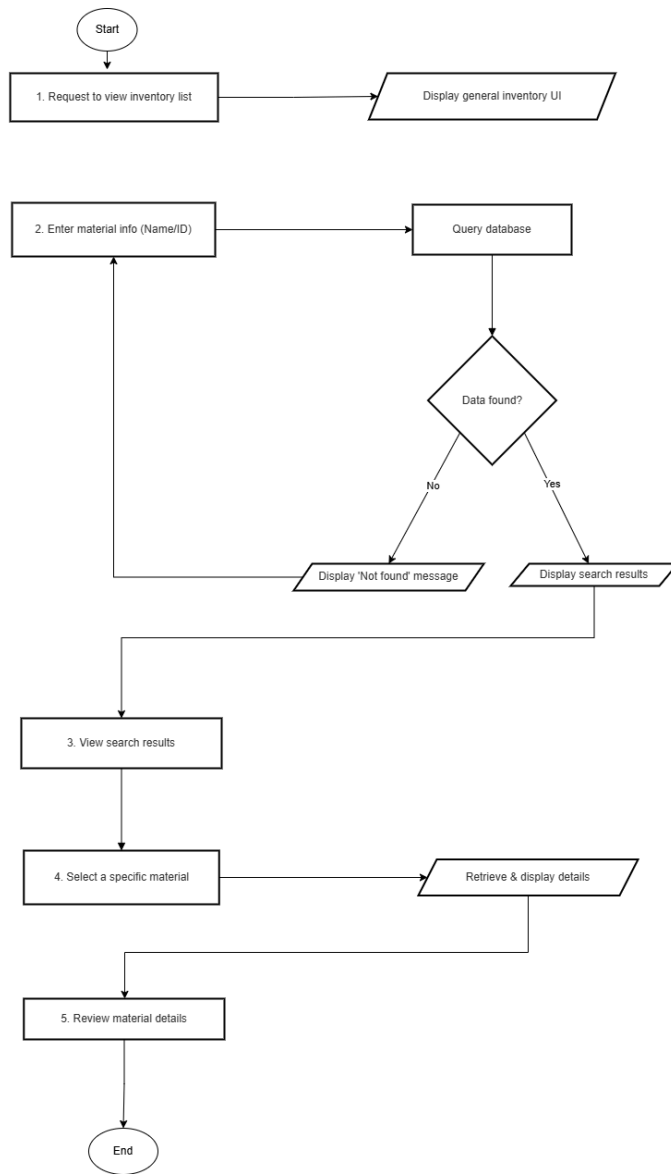
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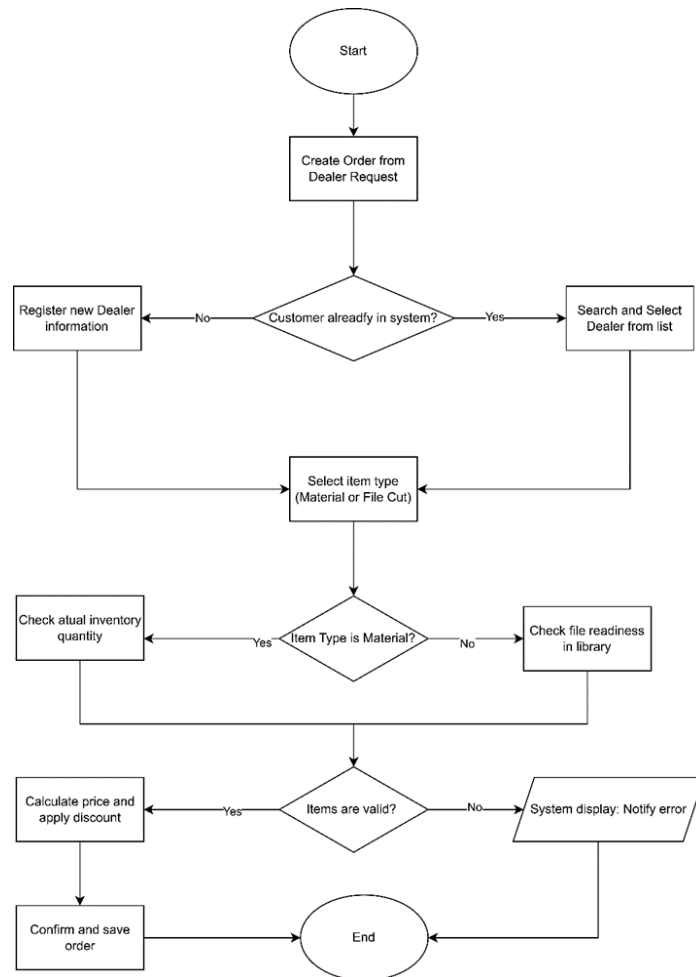
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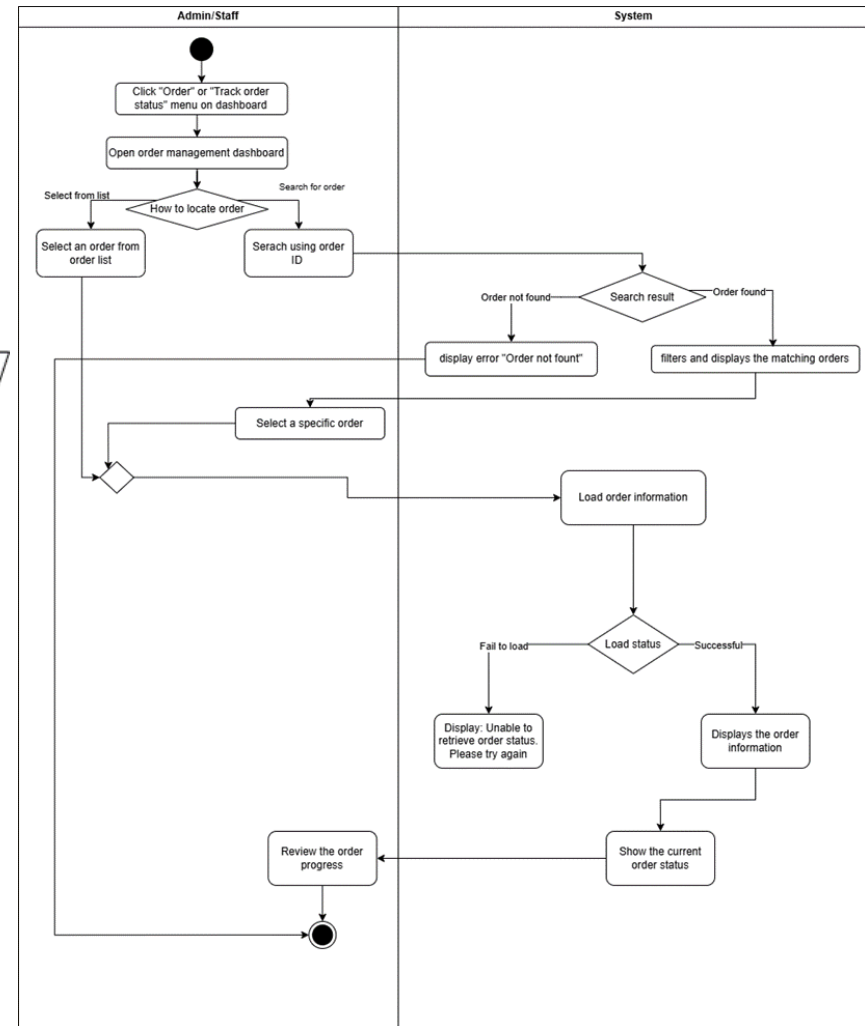
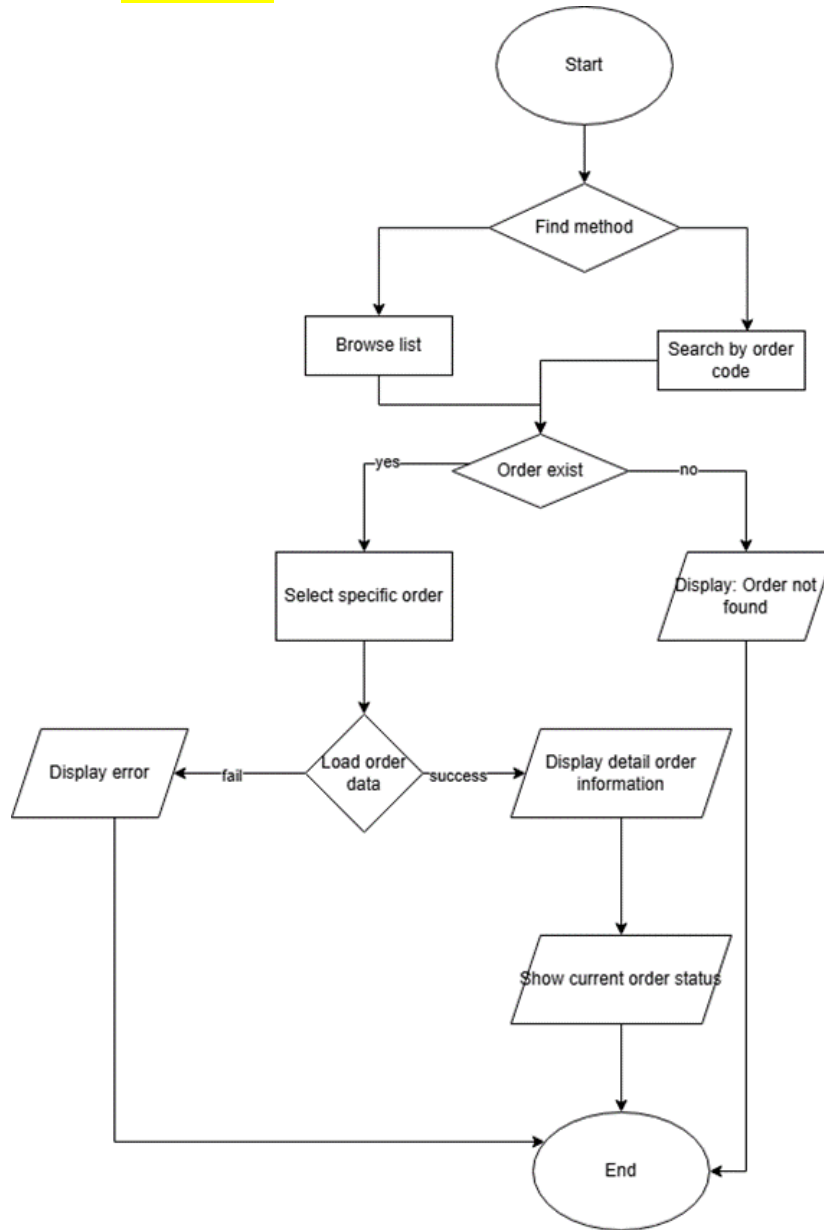
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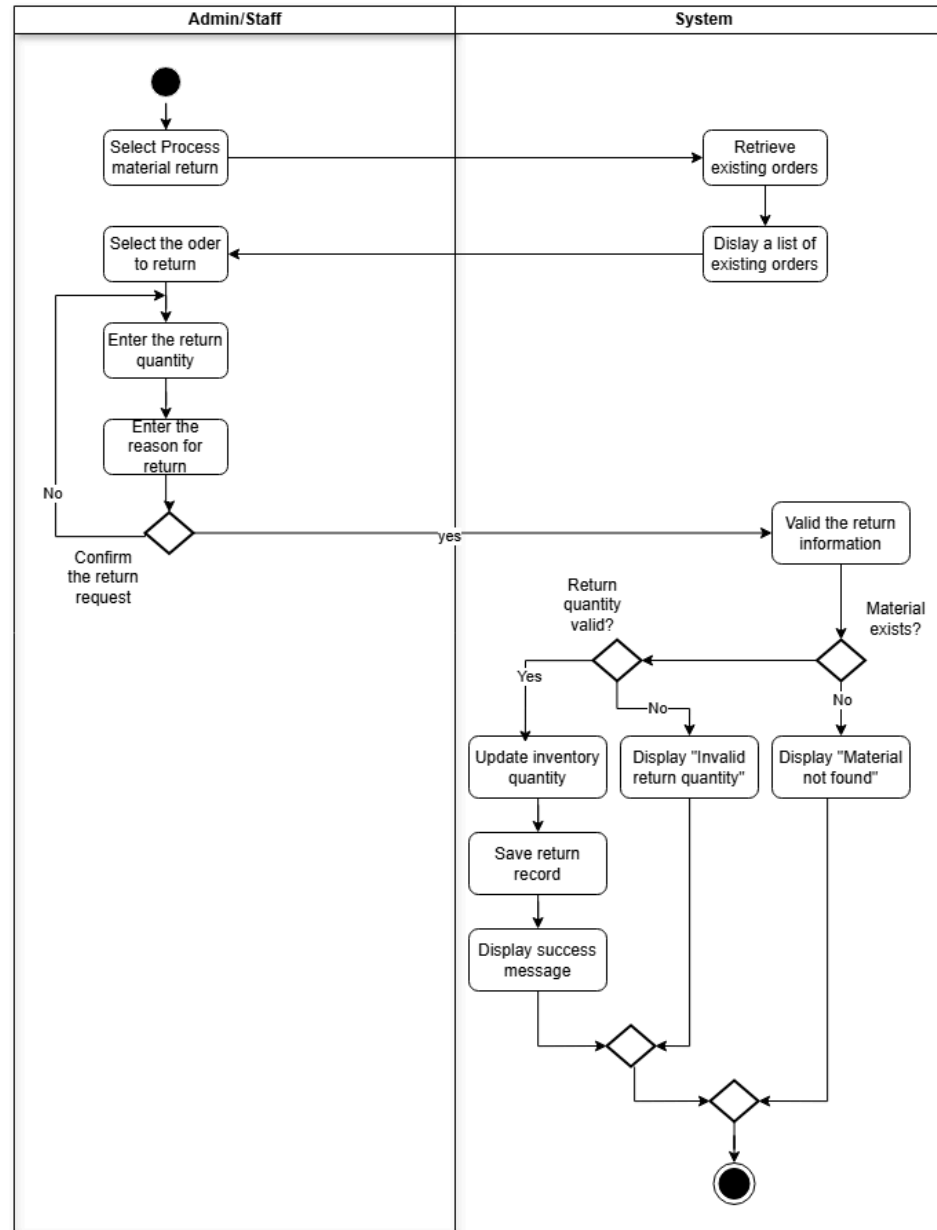
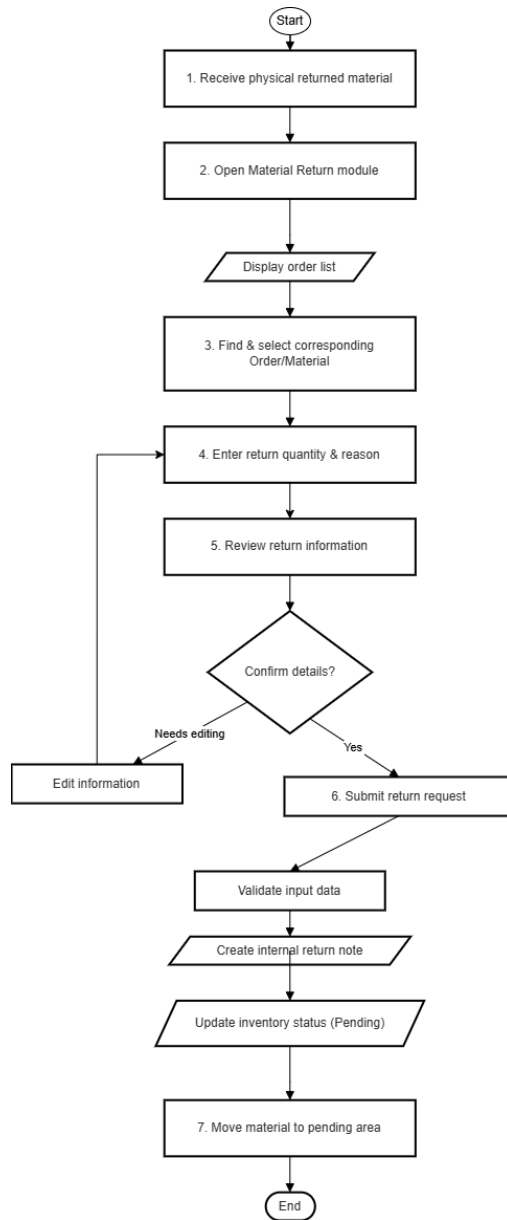
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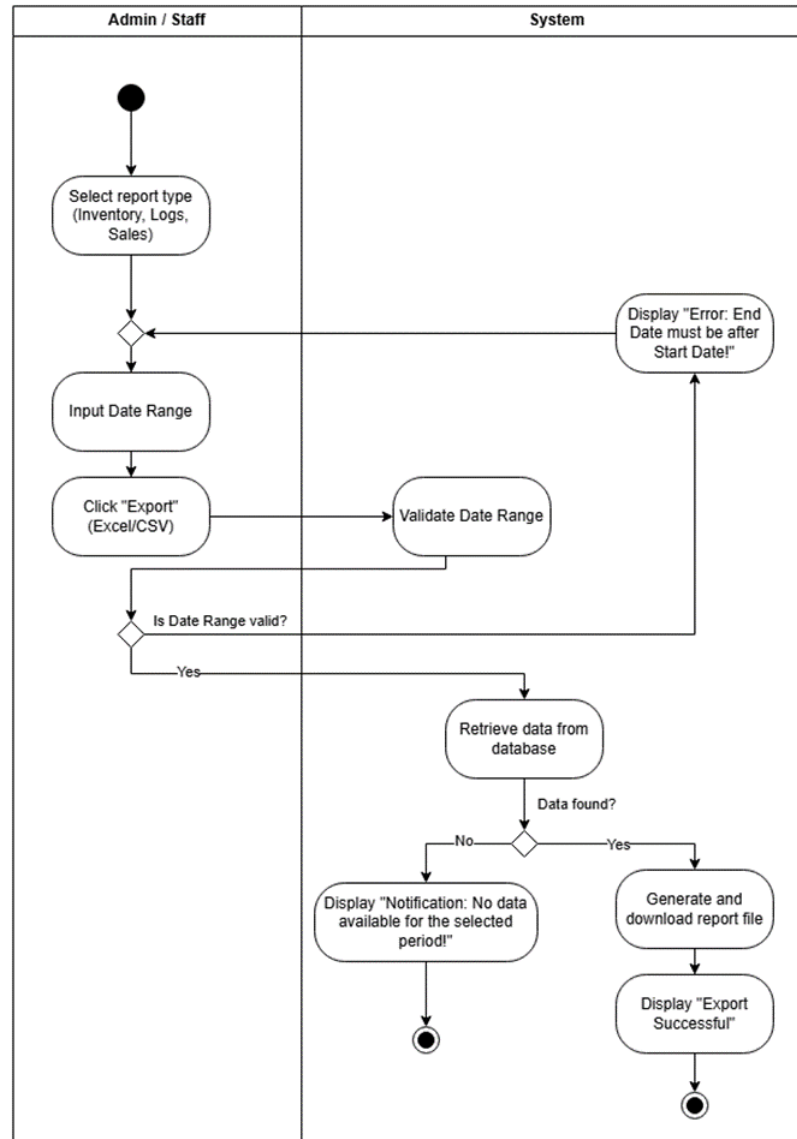
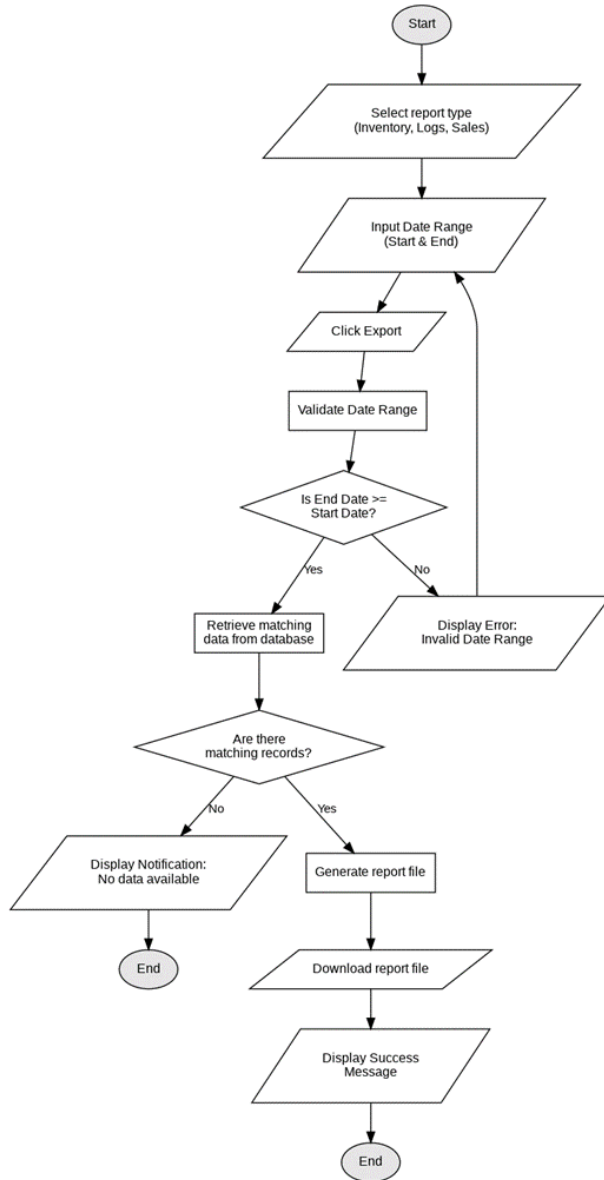
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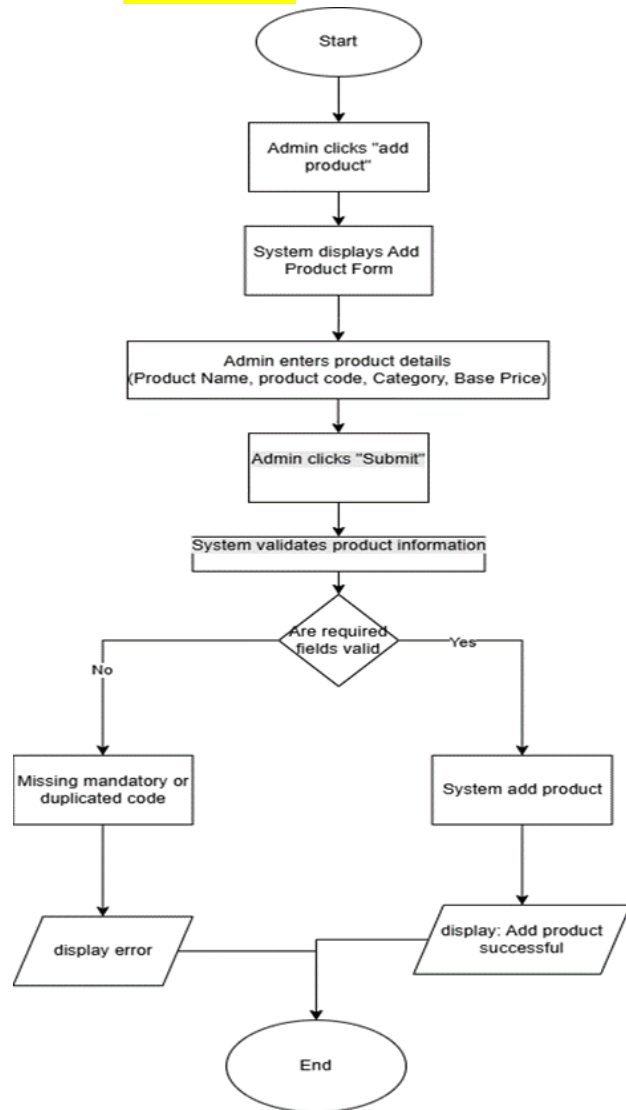
Use Case 13

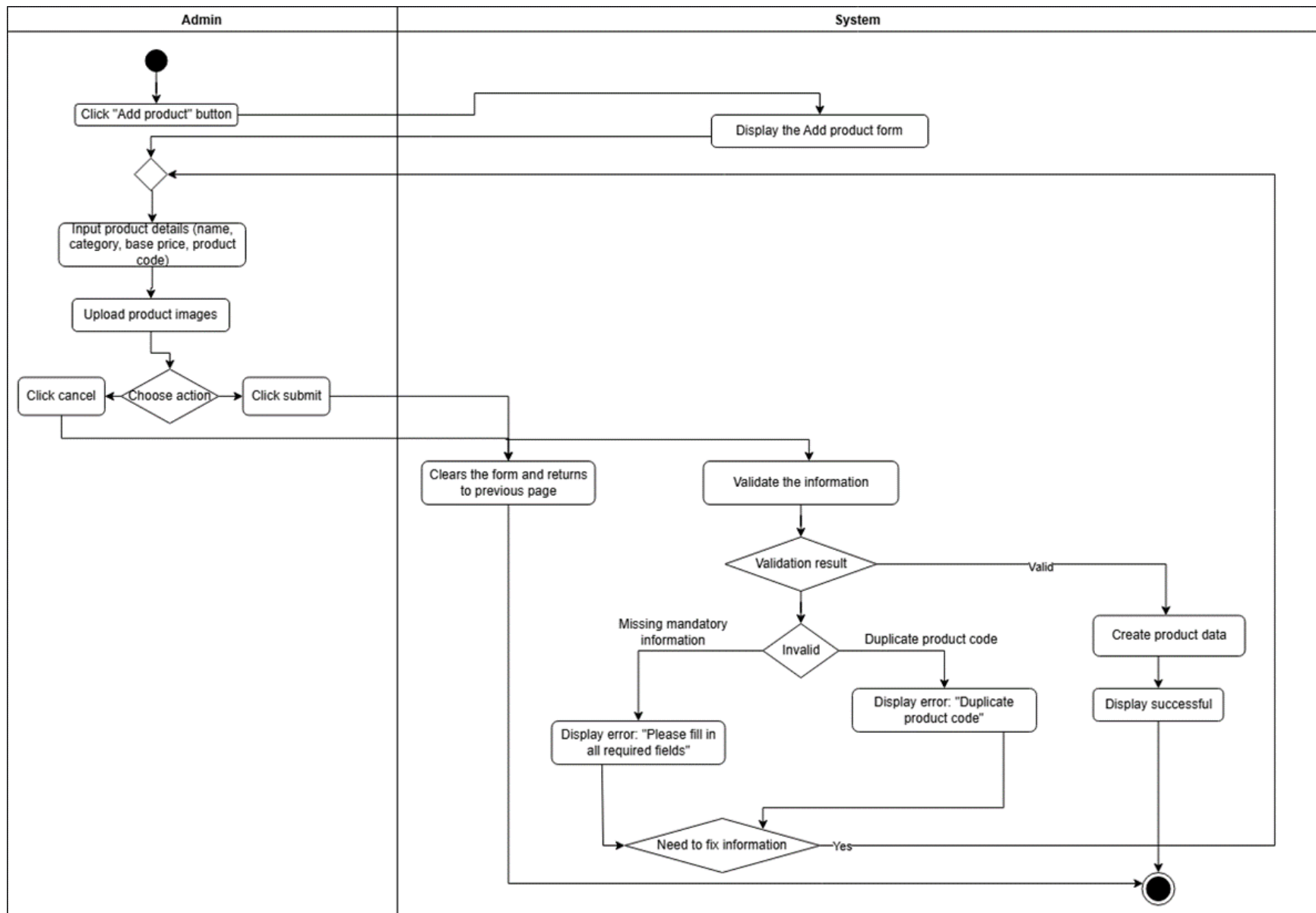


Use Case 14



Use Case 15





Appendix B – Business Rules Catalog

[Business case]

Name: Pwrap - Wrap Materials and Digital Patterns
Management System

Date: 02/04/2026

1. EXECUTIVE SUMMARY

The Pcut project aims to replace Pwrap's currently fragmented and manual processes with a centralized, integrated solution to manage both physical wrap materials (PPF, decals) and highly sensitive digital assets (cut files/patterns). The system will synchronize physical inventory tracking with a secure digital library, streamline B2B wholesale order fulfillment, and enforce strict role-based access. Key benefits include the elimination of inventory discrepancies, robust protection of the company's intellectual property (IP) against unauthorized leaks, and enhanced operational speed. The recommended solution involves developing a custom management system tailored to Pwrap's unique dual-asset business model, ensuring scalability and maximum data security.

2. PROBLEM STATEMENT

Current operations rely on manual, uncoordinated methods where physical stock and digital assets are managed separately, leading to:

- **Inefficiencies:** Time-consuming workflows and operational delays. Staff and technicians spend excessive time manually matching physical blank material rolls with the correct digital patterns. This disconnection reduces productivity and increases customer wait times.
- **Errors:** Inaccurate inventory tracking. Tracking physical wrap materials (in linear meters) manually or through basic spreadsheets is prone to errors. This leads to inaccurate stock levels, unexpected material shortages during production, and material waste.

- **Security & Compliance Risks:** Vulnerable digital assets. Without a centralized security layer, Pwrap's unique vector cut files (.cdr, .eps)—which are the company's core competitive advantage—are highly vulnerable to unauthorized access, internal theft, and external leaks.
- **Poor Visibility:** No real-time insights into wholesale operations. Management lacks real-time data on B2B Dealer order statuses, inventory depletion rates, and file download logs, hampering the ability to scale wholesale operations professionally.

3. ANALYSIS

- **What:** The Pcut system will automate inventory tracking (linear meters/rolls), secure digital cut file management (version control, watermarked previews, expiring download tokens), and streamline B2B order processing and returns.
- **Why:** Manual methods are unsustainable and pose a severe threat to the company's IP. Disconnected systems lead to high administrative burdens and inventory shrinkage.
- **Impact of Inaction:** Continued operational bottlenecks, high risk of proprietary digital patterns leaking to competitors (which could severely impact market position), and loss of B2B clients due to slow or inaccurate order fulfillment.
- **Resources Needed:** (Estimated)
- **Budget:** \$40,000 - \$60,000 (Software development, secure cloud infrastructure, database setup).
- **Timeline:** 2 months (Core Management MVP), 3 months (Full B2B Order Rollout).
- **Team:** Internal stakeholders (Sponsor, Admin) + Development Team (PM, BAs, UI/UX, System/Data Analysts, Developers).

4. SOLUTION OPTIONS

Option	Pros	Cons
1. Custom System (Pcut)	Perfectly tailored to the unique physical-digital hybrid model; high IP security (tokens/watermarks); full ownership.	Higher upfront development cost; requires internal testing and deployment time.
2. Off-the-Shelf ERP	Faster deployment; standard inventory features out of the box.	Cannot handle the specialized security needed for vector cut files; requires clunky

		workarounds; ongoing subscription fees.
3. Do Nothing	No upfront financial cost; no disruption to current habits.	Severe IP vulnerability; inventory errors persist; inability to scale wholesale operations.

4. COST BENEFITS ANALYSIS

Option	Cost	Benefits	ROI Timeline
Custom System	\$50,000	\$35K/year savings (efficiency/reduced waste) + Unquantifiable IP protection value.	1.5 years
SaaS Solution	\$15K/year	\$10K/year savings (inventory tracking only, digital files still at risk).	N/A (ongoing cost)
Do Nothing	\$0	Continued operational losses and risk of a massive financial hit if IP is stolen.	N/A

Key Benefits:

Digital Asset Protection: Eliminates unauthorized downloads via expiring tokens and granular role-based access, securing the company's competitive edge.

Inventory Accuracy: Real-time stock deductions (linear meters) combined with automated low-stock alerts prevent supply chain disruptions.

Time Savings: Streamlines the B2B order workflow from creation to fulfillment, significantly reducing administrative workload.

- RECOMMENDATION

Recommended Solution: Develop the custom Pwrap Management System (Pcut) with a phased implementation approach:

- **Phase 1:** Core Digital Library (secure upload, versioning, search, watermark previews) and Inventory Management (restock, view alerts).
- **Phase 2:** B2B Order Management (create orders, process returns, track status) and System Reporting.

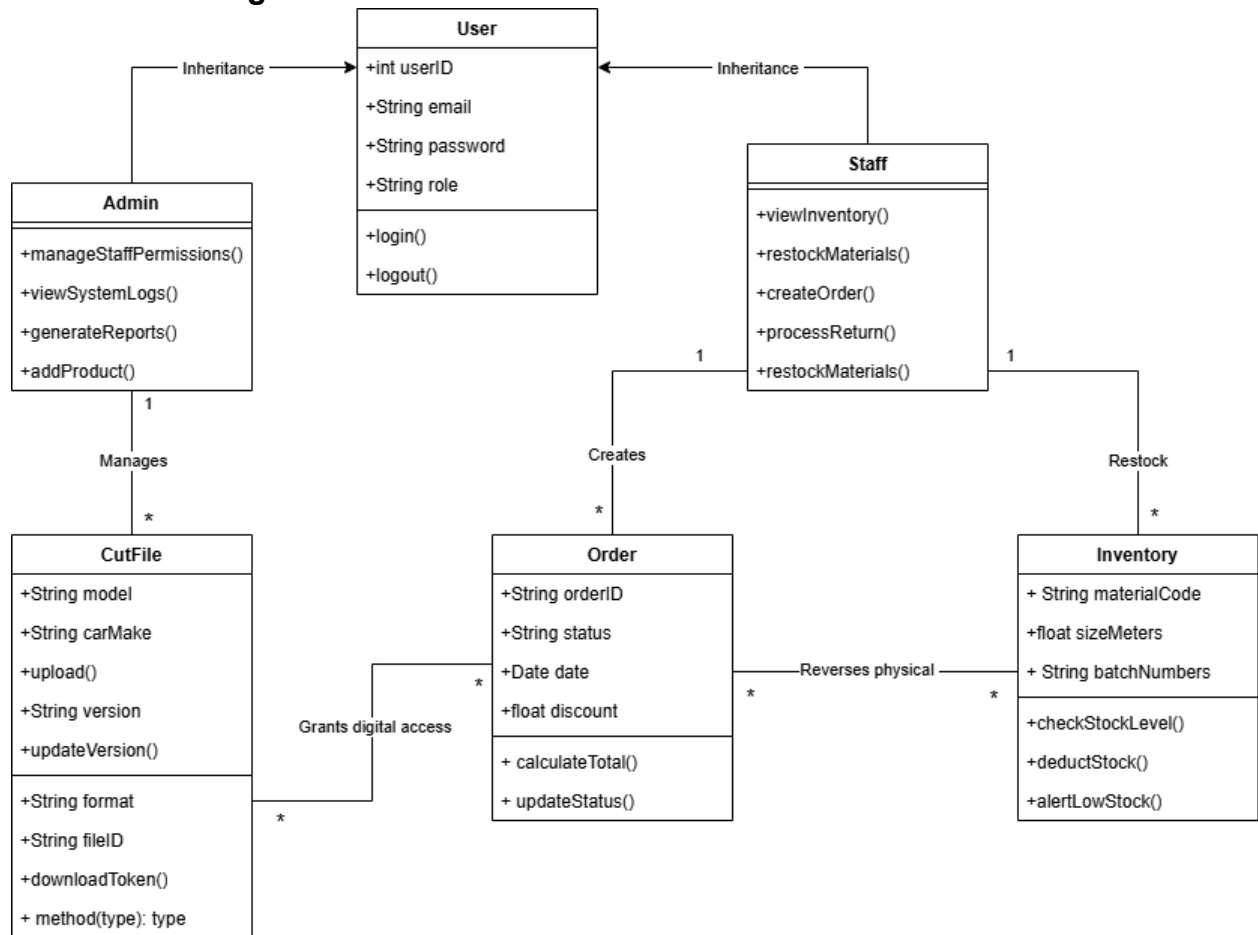
Justification:

Pwrap's proprietary digital patterns are its most valuable asset. Generic off-the-shelf software cannot provide the specialized security mechanisms (like temporary download tokens and watermarked vector previews) seamlessly combined with physical linear-meter inventory tracking. The long-term protection of IP and operational cost savings far outweigh the initial custom development investment.

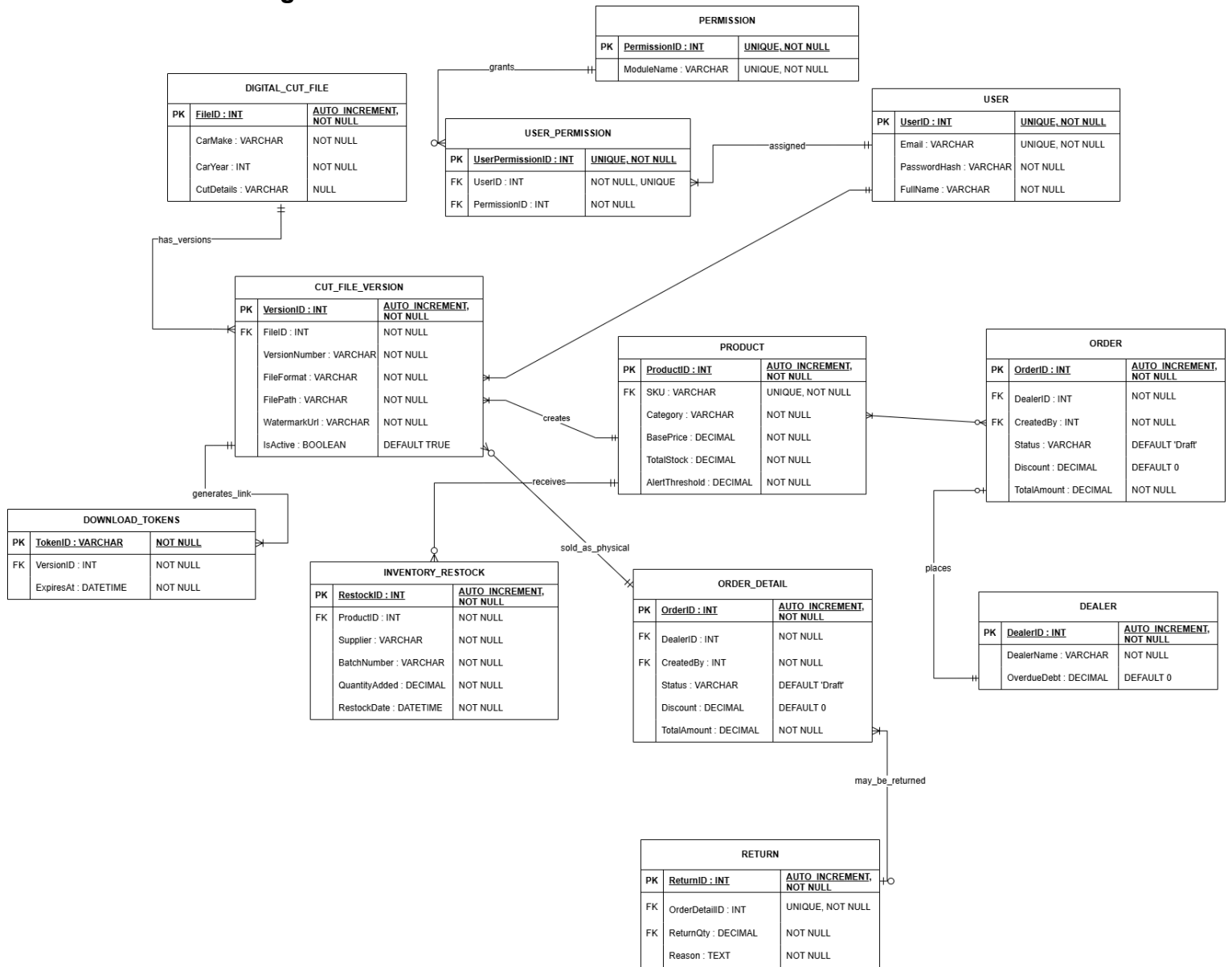
Next Steps: Secure final approval for the Business Requirements Document (BRD) from the Project Sponsor (Mr. Dao Duy Phong), allocate the project budget, and transition into the system design and development phase.

Appendix C- Models

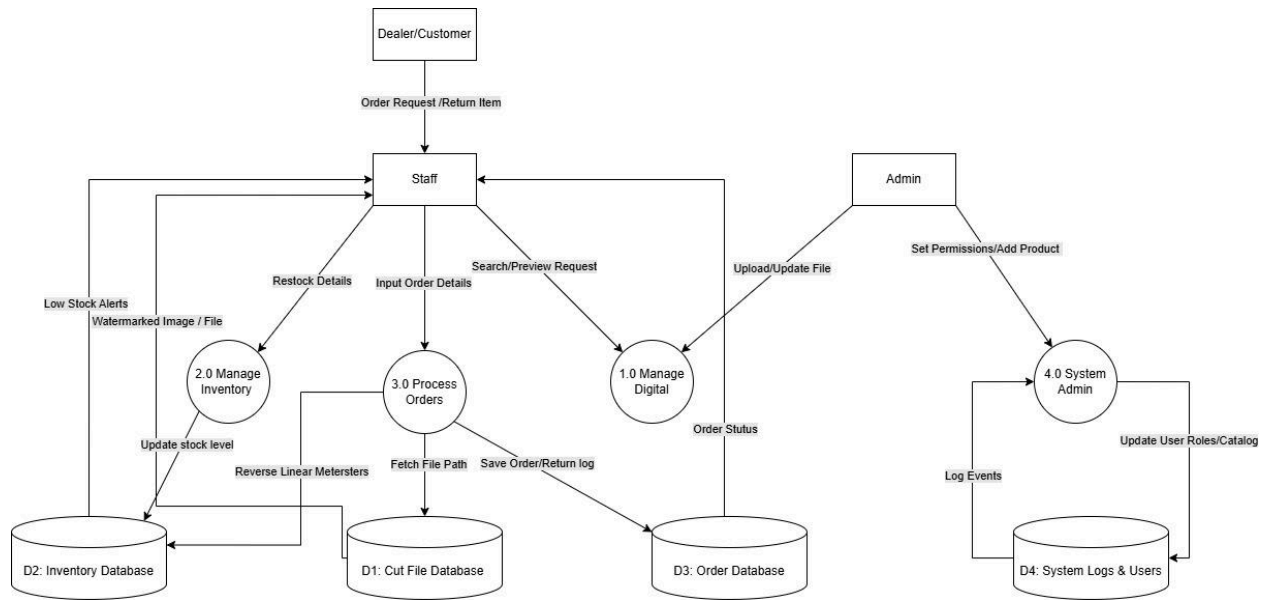
1. Class Diagram



2. ER Diagram



3. Data Flow Diagram



Use Case Narrative Instructions